Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score*

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Abstract

We propose a deployable downside-risk engine and a theory of when it pays. The engine — *amortized residual-hybrid* forecasting — keeps the GARCH conditional scale that industry models get right and replaces only the fixed innovation shape with a state-conditioned nonparametric quantile model trained *once, across hundreds of names*, finished with an extreme-value tail and a conformal recalibration. Three results. *First, amortization:* one fit replaces per-asset estimation, transfers to names never seen in training, beats own-history benchmarks at every listing age (6–10% of pinball loss for young listings), and prices assets with *no return history* — day-one risk numbers for IPOs that no per-asset model can produce. *Second, the engine beats what banks run:* against historical simulation, RiskMetrics, GARCH-t, GJR-skew-t, and filtered historical simulation on 140 CRSP names, it wins on average loss with date-level Diebold–Mariano significance (statistics 4.4–8.3), delivers the best-calibrated 97.5% Expected Shortfall (removing an 8–11% ES over-statement that overcharges capital), and with its tail stages its exception counts are consistent with nominal at both regulatory levels, a finding that survives per-asset and common-date-dependence restatements and holds through a 2008-crisis test window under the frozen specification; the edge grows at the ten-day horizon and holds in the 2020–22 stress window. *Third, a misspecification frontier says when the edge is large:* a real-time score (the excess kurtosis and asymmetry of recent GARCH-standardized residuals) concentrates the gain (+2.7%, DM 6.5, in its top decile; 12–14% on hyperinflation currencies), maps it across US equities, FX, twenty-six country indices, and forty-three cross-asset instruments, and replicates on an untouched 2000–2013 holdout run under registered predictions with the specification frozen (top decile +0.9%, DM 2.2; all other deciles indistinguishable from zero); the top-decile cells of all three signals survive Romano–Wolf familywise control over the full grid, strict calendar-time splits (including training that ends before the 2008 crisis), and a point-in-time universe that keeps delisted names — where the edge is larger, not smaller (top bucket +2.9%, DM 6.6). Where the score is quiet the engine ties the parametric benchmark, so the deployment rule is to run the engine everywhere and use the score as a model-risk monitor.

Keywords: Value-at-Risk; Expected Shortfall; FRTB; quantile regression; extreme value theory; conformal prediction; misspecification; amortized estimation.

*All results are out-of-sample under the walk-forward protocol described in the text; code and data availability are stated at the end.

JEL classification: C14; C22; C52; C58; G17; G28.

1 Introduction

Two communities measure conditional downside risk. Econometrics builds parametric filters — GARCH and its leverage and skew extensions — whose conditional scale dynamics are excellent and whose innovation law is fixed by assumption.[1] Machine learning builds distribution-free conditional quantile estimators — gradient-boosted quantile trees, neural implicit quantile networks (IQN), and, most recently, generative Bayesian computation (GBC) in the sense of Polson and Sokolov (2023) — which assume nothing about the innovation law but must learn everything from data. Industry practice, meanwhile, is neither: under the Basel Fundamental Review of the Trading Book (FRTB), the regulatory objective is Expected Shortfall at the 97.5% level, the workhorse internal models are historical simulation (HS) and GARCH-filtered historical simulation (FHS), and backtesting proceeds through VaR exception counts at 99% and 97.5%. Any claim that a new method "beats the industry standard" must therefore beat HS and FHS on $\text{ES}_{97.5}$ under exception-test discipline; beating a vanilla GARCH on average loss is not enough.

One distinction, stated before anything else, prevents the most common misreading of our results. When the misspecification score is quiet — roughly ninety percent of asset-days — the engine we deploy and the parametric benchmark are statistically indistinguishable: *a tie, not a loss.* The engine's worst outcome anywhere in our tests is that tie, so holding it costs nothing while it waits for the days on which it wins. What does lose on quiet days is a configuration we study only as a cautionary boundary and never deploy: a bare per-name neural quantile model, which a unit-matched GJR-GARCH-t beats by 4.5–6.2% at every horizon (DM t between +12 and +18)[2] — the failure that dictates the engine's hybrid design. The contribution is thus a *predictive theory of when flexible-shape methods win*, an engine that converts the theory into a risk model that meets and beats the industry standard on distributional accuracy while never doing worse than it on that yardstick — its one measured concession, the coverage shift's joint-score cost at 2.5%, is priced rather than hidden (Section 6) — and a demonstration that the generative members of the family deliver capabilities no parametric point-forecasting model offers at all.

[1] For readers outside econometrics: GARCH (generalized autoregressive conditional heteroskedasticity; Engle, 1982; Bollerslev, 1986) captures the most robust fact about markets — volatile days cluster — by letting today's variance be a weighted average of yesterday's variance and yesterday's squared surprise (Eq. 2). "Leverage" extensions (GJR; Glosten et al., 1993) let down-moves raise volatility more than up-moves; "skew-t" versions allow an asymmetric heavy-tailed innovation. What all of them fix in advance is the *shape* of the standardized surprise — the assumption this paper's score monitors.

[2] Throughout, DM denotes the Diebold and Mariano (1995) test: a t-statistic on the mean difference of two forecasters' losses (here, pinball loss), with a serial-correlation-robust standard error. Positive DM t favors the first-named model; $|t| > 1.96$ is significant at 5%.

1.1 Thesis

A single measurable quantity — the local misspecification of the parametric conditional tail, operationalized as the excess kurtosis and asymmetry of recent GARCH-standardized residuals — governs when distribution-free quantile methods beat industry-standard parametric VaR/ES. Where that misspecification is high the deployed engine wins decisively and with statistical significance; where it is low the engine and the parametric model are statistically indistinguishable — a tie, not a loss. Everything else in the paper is either evidence for this claim or the engineering that makes the engine deployable at industry standards.

1.2 Contributions

1. **The misspecification frontier** (Section 5). A live, per-asset, per-day score that predicts when nonparametric quantiles beat GARCH-t, with per-date Diebold–Mariano significance, validated across four universes: CRSP US equities (top-decile edge +2.71%, DM 6.54), FX (hyperinflation tier pooled DM 4.12), twenty-six country indices (developed→frontier gradient, corr 0.53), and forty-three cross-asset instruments (corr 0.43), with Korea-2026 as a sharpening negative control: price crashes are not residual misspecification.

2. **An industry-standard-and-above risk engine** (Section 6). The residual-hybrid — GARCH conditional scale × state-conditioned nonparametric residual quantile — with an extreme-value left tail and split-conformal recalibration: co-best with CAViaR[3] in the 90% Model Confidence Set[4], significantly better than GARCH-t/FHS/EWMA/HS/GJR-skew-t, best $\mathrm{ES}_{97.5}$ calibration, exception counts consistent with nominal at both regulatory levels (Kupiec+Christoffersen[5] at 99% via the EVT tail; conformal recalibration at 97.5%) — a finding that survives per-asset and date-clustered restatements and a 2008-crisis window rerun — the lowest joint $(\mathrm{VaR}, \mathrm{ES})$ FZ0 score at both levels on the full panel for the engine's accuracy layer, with the conformal shift's sharpness-for-coverage cost at 2.5% quantified rather than hidden, and an edge that grows at the FRTB ten-day horizon and holds in the 2020/2022 stress window.

3. **Amortization and the cold start** (Sections 3 and 7). The deployed forecaster is a single amortized model trained across hundreds of names: one fit replaces per-asset estimation, transfers to names the model has never seen, beats own-history benchmarks at *every* listing age (the advantage largest for young listings), and prices assets with no return history at all — the one regime where it is the only feasible model.

[3] CAViaR (Engle and Manganelli, 2004) models the quantile itself as an autoregressive process estimated on the pinball loss — the strongest non-nested rival in the battery.

[4] The MCS (Hansen et al., 2011) is the subset of models that cannot be statistically rejected as best at a given confidence level; "co-best" means surviving the elimination sequence.

[5] Kupiec (1995) tests whether the *number* of days losses exceed VaR matches the nominal rate; Christoffersen (1998) additionally tests that exceptions arrive *independently* rather than in clusters. Both are standard in regulatory backtesting.

4. **A theory of where the edge lives** (Section 3, proofs in Appendix A). A pinball regret identity showing that a forecaster's excess loss is quadratic in its quantile wedge, so any edge must concentrate where the wedge is largest, and a tail-wedge proposition showing that innovation-shape misspecification creates that wedge in the tail — together the reason the frontier is where the empirical edge must, and does, appear.

Three neighboring questions are out of scope and claimed nowhere in this paper: posterior uncertainty bands on the reported (VaR, ES); likelihood-free amortized posteriors for parametric simulators; and the multivariate portfolio tail. This is a point-forecasting paper, and it is self-contained: every stage the deployed engine uses is developed, and every formal result it relies on proved, here.

Summary of results. Table 1 states every headline magnitude with its test statistic. Panel A is the presented engine's record — its worst cell on the accuracy yardstick is a *tie*, and its one scored concession — the coverage shift's 2.5% joint-score cost — is flagged in its own row rather than absorbed into an average. Panel B is not the engine's record at all: it collects the boundary evidence — configurations we deliberately do *not* deploy and negative controls — that motivates the engine's design and delimits where the flexible-shape edge lives.

Two pieces of context make these magnitudes readable. First, the *asymmetry of the loss profile* is the economic argument for adoption: when the frontier is quiet, the deployable engine ties or edges the parametric benchmark (its calm-quintile edge is ≈ 0 but not negative, and by construction it nests FHS); the one configuration that loses materially — a bare standalone neural network on single names, -4.5 to -6.2% — is the configuration Table 2 says never to deploy. Adopting the engine therefore risks a rounding error to collect a concentrated +2.7% in the decile where risk models actually fail, and 12–14% where the parametric assumption breaks entirely. Second, for readers without a pinball intuition: a percentage pinball edge is a percentage reduction in the average forecast error at the quoted level, and its capital-relevant translation is the 8–11% $ES_{97.5}$ over-statement by FHS/GARCH that the engine removes — roughly that fraction of a desk's market-risk capital charge held against a risk that is not there.

Throughout, the exposition follows three reporting conventions fixed in advance. First, the strongest tabular estimator[6] in the family is gradient-boosted trees, not the neural network; we present GBC as the framework, trees as the strongest estimator, and the tail-aware neural IQN as the canonical generative realization, competitive after tuning and uniquely able to sample. Second, residual kurtosis is necessary but not sufficient for a nonparametric win (carbon and corn are high-kurtosis ties; Baltic freight wins at moderate kurtosis through limit-move dynamics); the frontier is a strong regularity with counterexamples, and we show them. Third, every negative result that shaped the design — the retired same-universe artifact, the electricity log-return artifact, the failed hierarchical blends, the failed learned gate, the three multivariate losses — is reported in the text.

[6] "Tabular" data: observations that are rows of engineered features (here, the state vector s_t) rather than raw sequences or images — the setting where tree ensembles remain the strongest general-purpose learners.

2 Background and industry benchmark

2.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t, the level-α Value-at-Risk is the conditional quantile $\mathrm{VaR}_t^\alpha = F_t^{-1}(\alpha)$ and the Expected Shortfall is $\mathrm{ES}_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq \mathrm{VaR}_t^\alpha]$. FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at $\mathrm{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket) while backtesting remains VaR-exception based: more than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a desk off the internal-models approach. The empirical bar for "beating the industry standard" is therefore joint: accuracy on $\mathrm{ES}_{97.5}$ and pinball, *and* unconditional (Kupiec, 1995) (the Kupiec test: are there the right number of VaR breaches?) plus independence (Christoffersen, 1998) (the Christoffersen test: are breaches independent, rather than clustered in time?) exception tests at both levels, with significance assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) (a t-test on the difference in loss between two forecasts) and the Model Confidence Set (Hansen et al., 2011) (the set of models statistically indistinguishable from the best one).

2.2 The parametric family, the simulation family, and CAViaR

Our battery spans the models banks actually run: historical simulation; EWMA/RiskMetrics (Zumbach, 2007) (an exponentially weighted moving average of squared returns, giving recent days more weight); GARCH(1,1)-t (Bollerslev, 1986); GJR-GARCH-skew-t (adding leverage and skewness; Glosten et al., 1993); GARCH-FHS (parametric scale, empirical standardized residuals — the volatile-regime best practice; Barone-Adesi et al., 1999); and SAV-CAViaR (Engle and Manganelli, 2004), included after an anti-strawman audit showed its omission flattered our model. CAViaR (conditional autoregressive Value-at-Risk) skips modeling the distribution entirely and lets the quantile itself follow a recursion estimated on the pinball loss; the SAV (symmetric absolute value) specification is $q_t(\tau) = \beta_0 + \beta_1\, q_{t-1}(\tau) + \beta_2\, |r_{t-1}|$ — today's VaR is yesterday's VaR adjusted by the size (not sign) of yesterday's move, making it the semiparametric analogue of a GARCH recursion written directly on the quantile.

2.3 Distribution-free conditional quantiles and GBC

The closest antecedent in this journal's own pages is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names — evidence of a shared volatility mechanism that our amortized design also exploits. The differences delimit this paper's contribution. Their target is a point forecast of realized variance; ours is the full conditional return distribution and its regulatory tail, held to calibration, exception-test, and capital discipline. Their question is whether machine learning wins on average; ours is *when* — and our answer, that on roughly ninety percent of asset-days it does not, reads as a caution against extrapolating average ML superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at every listing age. And we reach the opposite estimator verdict on tabular state (trees over networks), a divergence we attribute to the different target and feature set rather than a contradiction.

The nonparametric family estimates $Q_\theta(\tau \mid x_t)$ directly by pinball-loss empirical risk minimization (Koenker and Bassett, 1978): gradient-boosted quantile trees (GBM) and the implicit quantile network of Dabney et al. (2018) (IQN: the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). Casting GBC as ERM conditional-quantile regression on observed pairs removes its forward-simulator requirement and extends it to observational time series; the generative reading — draw $\tau \sim U(0, 1)$, output $Q_\theta(\tau \mid x)$ — is noted as a capability only; sampling and posterior uncertainty are outside this paper's scope. Generalized Bayes (a posterior built from a loss function in place of a likelihood) enters through the Gibbs posterior $\pi_n(\theta) \propto \exp\{-\omega \sum_t \ell_\theta(z_t)\}\pi(\theta)$ — a posterior built from a loss rather than a likelihood, targeting the risk minimizer directly (Jiang and Tanner, 2008; Bissiri et al., 2016); distribution-free finite-sample (exchangeability-based) guarantees enter through split-conformal and adaptive conformal inference (recalibrating predicted intervals on held-out data so that coverage holds without distributional assumptions) (Gibbs and Candès, 2021).

3 Methods

This section is written to be self-contained: a reader from any one of econometrics, machine learning, or risk management should be able to reimplement every estimator from the formulas and algorithm boxes below. (For a finance-journal submission the algorithm boxes and the proof of Proposition 1 can migrate to an appendix unchanged; they are self-contained.) A cross-field glossary of acronyms and jargon is provided in the Online Appendix.

3.1 Notation and objects

- r_t — return of an asset on day t; $\mathcal{F}_{t-1}$ — information available before t.
- $Q_t(\tau)$ — the conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$, formally $Q_t(\tau) = \inf\{q : \Pr(r_t \le q \mid \mathcal{F}_{t-1}) \ge \tau\}$: the number the return falls below with probability τ. Value-at-Risk is its negative, $\mathrm{VaR}_t^\tau = -Q_t(\tau)$; Expected Shortfall is the average of the quantile curve beyond it, $\mathrm{ES}_t^\tau = -\tau^{-1} \int_0^\tau Q_t(u)\, du$ — the expected loss on the days VaR is breached.
- $z_t = (r_t - \mu_t)/\sigma_t$ — the *standardized residual*: the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law.
- s_t — the state vector (trailing realized volatility, residual-shape scores, lags) on which nonparametric quantiles condition.

- $\tau \sim \lambda$ — a quantile level drawn from a base distribution λ; $H_\phi(s, \tau)$ — a learned generator mapping (state, level) to a quantile, in the sense of Polson and Sokolov (2023): $z \stackrel{D}{=} H(s, \tau)$, $\tau \sim U(0, 1)$, is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) \;=\; u\,(\tau - \mathbb{I}\{u < 0\}), \qquad u = z - q, \tag{1}$$

whose expectation is uniquely minimized by the true conditional quantile (Koenker and Bassett, 1978): differentiating $\mathbb{E}\,\rho_\tau(z-q)$ in q and setting it to zero gives $\Pr(z < q) = \tau$, so the loss *elicits* the quantile the way squared error elicits the mean — under-prediction is charged τ per unit, over-prediction $1 - \tau$, and the charges balance exactly at the τ-quantile. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once: the 1-Wasserstein distance between two laws is the L_1 distance between their quantile functions, $W_1(F, G) = \int_0^1 |F^{-1}(\tau) - G^{-1}(\tau)|\,d\tau$, so pinball empirical-risk minimization across levels is distribution matching in W_1 — density-free, which is what permits every estimator below to dispense with a likelihood.

Why this loss and not another. Three alternatives are standard, and the choices are deliberate. (a) The continuous ranked probability score (CRPS) is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F, y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\,d\tau$ (Gneiting and Raftery, 2007); scoring by CRPS therefore weights the distribution's body and tails equally by construction. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail — CRPS would average those differences away against a body where all competitors agree. We therefore score pinball *at the levels that are decision-relevant*, and use tail-weighted τ-sampling in training for the same reason. (b) Expected Shortfall is not elicitable on its own — no loss function is minimized by the true ES alone — but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016); the FZ-scored re-evaluation of the battery is a registered extension (Section 8). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them.

3.2 The final model: the residual-hybrid engine

The deployable model of this paper is *not* a bare neural network; it is a four-stage composition, each stage keeping what an industry model gets right. The design principle that survives every experiment is a division of labor: *let the parametric filter do what it is good at (conditional scale), and spend the nonparametric capacity only on what it is bad at (the conditional shape of the innovation law).* Intuitively: GARCH answers "how big is a typical move today?" extremely well, and badly answers "given a typical move size, how likely is a move five times that, and is down likelier than up?" The engine keeps GARCH's answer to the first question and learns the second from data. Each finishing stage has the same one-line logic: the EVT tail says *beyond the data, extrapolate with the law extreme-value theory proves exceedances must follow, instead of trusting a fitted model outside its training range*; the conformal stage says *measure your own miss rate on data you have not touched, and shift the curve by that measured miss* — calibration by bookkeeping, no assumptions needed.

Why one should expect this to work before seeing data. The architecture is not an empirical accident; each piece is where prior reasoning puts it. Conditional *scale* is the strongest, best-understood signal in daily returns (volatility clustering is the most robust stylized fact in the field), and it is identified by a handful of parameters — so a parametric filter estimates it with minimal variance, and a nonparametric model that re-learns it merely spends data on what four parameters already buy. Conditional *shape*, by contrast, is where the parametric family is constrained by assumption and where per-asset data is too scarce to fit flexibly — tail events are rare by definition — so that is where nonparametric capacity earns its keep, and only if it can borrow strength across assets. This is classical bias–variance division of labor: parametric where the structure is known, flexible where it is not, which is why the hybrid should (and does) dominate both endpoints. Pooling residuals across names is in turn the standard hierarchical-shrinkage argument: once name-specific scale is divided out, standardized surprises are approximately draws from a common state-conditional law, and estimating a shared object from the whole cross-section improves on any single name's estimate whenever units are scarce individually and similar collectively — the same logic that makes partial pooling dominate in hierarchical Bayes. The EVT stage is backed by a limit theorem: exceedances over a high threshold converge to the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), so tail extrapolation by GPD is the principled choice for *any* body model. And the conformal stage's guarantee is Proposition 1, not an empirical observation. The experiments in this paper are therefore tests of a mechanism argued in advance, not a specification search rewarded after the fact.

Stage 1 (parametric scale). Fit GARCH-t,

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (standardized)}, \tag{2}$$

by quasi-maximum likelihood on a trailing window, and keep only $\widehat{\sigma}_t$ and the residuals $\widehat{z}_t = (r_t - \widehat{\mu})/\widehat{\sigma}_t$.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\widehat{\phi} \;=\; \arg\min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau\big(\widehat{z}_{j,t} - q_\phi(\tau \mid s_{j,t})\big). \tag{3}$$

Stage 3 (EVT tail). Beyond the estimation range of (3), extrapolate with the distribution extreme-value theory says exceedances must follow (Pickands, 1975; McNeil and Frey, 2000).[7]

[7] Extreme-value theory has two classical formulations, and the distinction matters for reading the literature. The *block-maxima* approach models the maximum of each block (e.g., yearly worst loss), which converges to the generalized extreme value (GEV) family — Gumbel, Fréchet, or Weibull type depending on the tail. The *peaks-over-threshold* (POT) approach, used everywhere in this paper, models all exceedances over a high threshold, which converge to the generalized Pareto distribution (GPD) (Balkema and de Haan, 1974; Pickands, 1975). POT uses the data more efficiently (every large loss contributes, not just one per block), which is why it is the standard in risk management (McNeil and Frey, 2000; Embrechts et al., 1997). The GPD's shape parameter ξ

On losses $X = -\widehat{z}$, fix a threshold u at the empirical $(1 - p_0)$ loss quantile with $p_0 = 0.025$, fit the generalized Pareto distribution (GPD) $G_{\xi,\beta}(x) = 1 - (1 + \xi x/\beta)^{-1/\xi}$ — shape ξ sets how heavy the tail is, scale β how wide — to exceedances $X - u > 0$ strictly on past data, and set, for $\tau \leq p_0$,

$$\widehat{q}^{\mathrm{EVT}}(\tau) \;=\; -\Big[u + \tfrac{\beta}{\xi}\big((\tau/p_0)^{-\xi} - 1\big)\Big], \tag{4}$$

which is continuous at its own endpoint, $\widehat{q}^{\mathrm{EVT}}(p_0) = -u$ (and reduces to $-[u + \beta \log(p_0/\tau)]$ in the $\xi \to 0$ exponential limit; requires $\beta > 0$ and $1+\xi(x-u)/\beta > 0$). Since u is the *unconditional* empirical loss threshold while the Stage-2 body quantile at p_0 is the *state-conditional* $\widetilde{q}(p_0 \mid s)$, the splice is exactly continuous only when the threshold is anchored at the body's own p_0-quantile, $u = -\widetilde{q}(p_0 \mid s)$; otherwise a small level shift of size $|-u - \widetilde{q}(p_0 \mid s)|$ remains at the join and the two pieces are spliced to agree in practice by construction. The tail is applied selectively where the raw tail under-covers (it *hurts* some volatility-index series and is not applied universally).

Stage 4 (conformal finish). On a held-out calibration split of size n, compute the signed errors $e_i = \widehat{z}_i - \widehat{q}(\tau \mid s_i)$ and shift the curve per level by the empirical correction c_τ chosen as the $\lceil (n{+}1)\tau \rceil$-th order statistic of $\{e_i\}$: $\widetilde{q}(\tau \mid s) = \widehat{q}(\tau \mid s) + c_\tau$ (Vovk et al., 2005; Romano et al., 2019). This distribution-free layer repairs the 97.5% level that every model in the battery otherwise fails.

Proposition 1 (split-conformal validity)**.** *If the calibration and test residuals $e_1, \ldots, e_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n{+}1)\tau \rceil \leq n$ the shifted quantile satisfies*

$$\tau \;\leq\; \Pr\big(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\big) \;=\; \frac{\lceil (n{+}1)\tau \rceil}{n+1} \;<\; \tau + \tfrac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of e_{n+1} among $\{e_1, \ldots, e_{n+1}\}$ is uniform on $\{1, \ldots, n+1\}$; the event $z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})$ equals $\{e_{n+1} \leq e_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \leq \lceil (n{+}1)\tau \rceil\}$, whose probability is $\lceil (n{+}1)\tau \rceil/(n+1) \in [\tau, \tau + \frac{1}{n+1})$ (Vovk et al., 2005). The full argument, the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1), and the boundary convention $c_\tau = +\infty$ when $\lceil (n{+}1)\tau \rceil = n+1$, are in Appendix A. The guarantee is marginal, not conditional on s_{n+1}. □

The proposition's scope is narrower than its use. In deployment the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable; the proposition therefore *motivates* the construction, while the evidence that it works here is empirical — the exception-test repairs of Section 6 (the conformal stage is the only reason any entrant passes Kupiec at 97.5%). Where regime drift erodes a static shift, an *adaptive* update (Gibbs and Candès, 2021) is the natural escalation; the engine's annual-refit schedule sits between the static and adaptive poles, and the exception tests are the arbiter.

corresponds to the GEV type: $\xi > 0$ is the heavy-tailed (Fréchet) case relevant for financial losses.

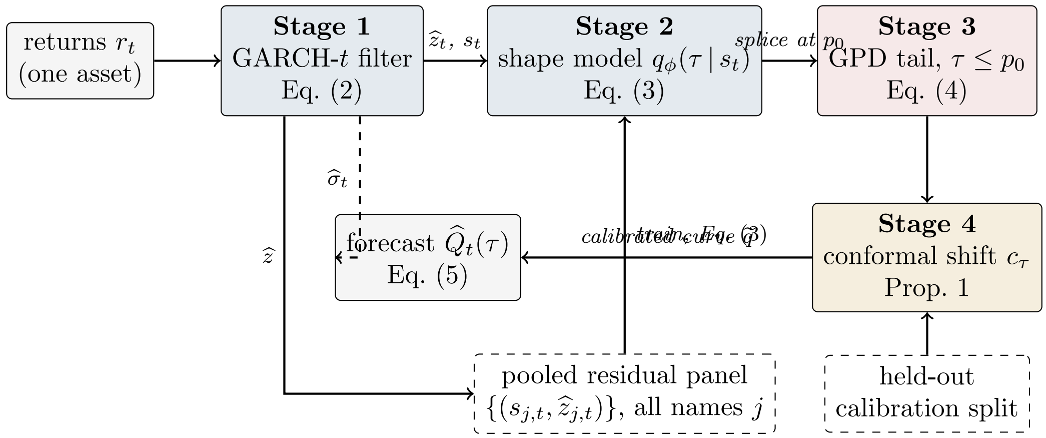

Figure 1: The residual-hybrid engine as a pipeline. Solid arrows carry data; the dashed top arrow carries the conditional scale $\widehat{\sigma}_t$ that rescales the finished residual quantile curve; dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal guarantee). Every stage refits on the walk-forward schedule using only past data.

The full forecast composes the stages:

$$\widehat{Q}_t(\tau) \;=\; \widehat{\sigma}_t \cdot \Big[\, \widetilde{q}_\phi(\tau \mid s_t)\, \mathbb{I}\{\tau > p_0\} \;+\; \widehat{q}^{\,\mathrm{EVT}}(\tau)\, \mathbb{I}\{\tau \le p_0\} \Big]. \tag{5}$$

Industry models as special cases. Equation (5) nests the standard toolkit. Historical simulation: $\widehat{\sigma}_t \equiv 1$ and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS): q the unconditional empirical quantile of past *residuals* — Stage 2 with the state dependence deleted. GARCH-t: $q = t_\nu^{-1}$, a fixed parametric shape. The hybrid is therefore "FHS and above": whatever FHS captures, (3) can represent, plus conditional shape. (CAViaR sits outside the nesting — it autoregresses on the quantile directly, with no scale/shape split — and is the one rival the engine only ties, Section 6.)

Algorithm 1 (residual-hybrid engine). (i) Fit (2) per asset on a trailing window; store $\widehat{\sigma}_t$, $\widehat{z}_t$. (ii) Pool $(s_{j,t}, \widehat{z}_{j,t})$ across names; train the shape model by (3). (iii) Fit the GPD tail (4) on past exceedances; splice at p_0. (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data. (v) Forecast by (5); refit all stages on the walk-forward schedule (annual refit; all windows trailing; evaluation strictly out-of-sample).

3.3 Estimating the shape: trees for accuracy, networks for generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3), and the paper is deliberate about which to use when (Table 2).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta\, h_m(\cdot)$ is grown stagewise, each new tree h_m fit to the negative gradient of the pinball loss (τ_k where the current model under-predicts, $\tau_k - 1$ where it over-predicts); the fitted grid is made non-crossing by monotone rearrangement — sorting the values $\{q(\tau_1|s), \ldots, q(\tau_K|s)\}$ in τ, which never increases the loss (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest *point* estimator — about 0.75% better pinball than the tuned network — consistent with the broader tabular-learning literature.

The neural implicit quantile network (IQN; the GBC estimator). A τ-conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s,\tau) \;=\; g\big(\psi(s)\circ\varphi(\tau)\big), \qquad \varphi_j(\tau) = \mathrm{ReLU}\Big(\textstyle\sum_{i=0}^{n_c-1}\cos(\pi i\tau)\,w_{ij} + b_j\Big), \tag{6}$$

where $\circ$ is elementwise multiplication and g, ψ are feed-forward nets. Three modifications, each forced by a documented failure, make it competitive on financial data: tail-aware τ-sampling, $\lambda = \frac{1}{2}\,U(0,1) + \frac{1}{2}\,\mathrm{Beta}(0.3, 0.3)$, oversampling the extremes that uniform sampling starves (the raw net's 99% breach rate was 3.2%); monotone rearrangement of the τ-curve; and the conformal layer of Section 3.2. The network's earning is not point accuracy but *generation*: with $\tau \sim U(0,1)$, $z^* = H_\phi(s,\tau)$ is an exact sampler of the fitted conditional law (the inverse-CDF representation) — a capability no tree ensemble or GARCH filter provides.

How training works, concretely. The objective is the empirical version of (3): mini-batch stochastic gradient descent on $\frac{1}{|B|}\sum_{(j,t)\in B}\rho_{\tau_{jt}}\big(\widehat{z}_{j,t} - H_\phi(s_{j,t},\tau_{jt})\big)$, where each example in each batch receives a *fresh* draw $\tau_{jt} \sim \lambda$ every epoch — over training the network sees every observation at many random levels, which is what lets a single set of weights carry the entire quantile curve. Early stopping is on held-out pinball loss; monotone rearrangement and the conformal shift are applied *after* training as post-processing; and the whole pipeline (filter, shape model, tail, shifts) is refit on the annual walk-forward schedule using only past data at every step. Exact layer sizes, learning rates, and seeds are recorded in the project's run configurations and reported with the code release rather than inline.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names — in generative-Bayes terms, H is an amortized map evaluated at any new (s,τ) without per-asset refitting. The intuition for why this works, and why it works at cold-start in particular, is actuarial. A per-asset GARCH learns *name-specific parameters* from that name's own history, so with no history it has nothing. The amortized model instead learns one *universal mapping from observable state to residual shape* — "assets whose trailing volatility looks like this, with characteristics like that, produce surprises shaped like so" — estimated from millions of asset-days of *other* names. A newly listed asset is not a new problem: its first ticks and characteristics place it at a point of state space already densely populated by other names' history, much as an insurer prices a new driver from the risk table built on millions of existing drivers rather than from the new driver's empty claims file. As the name's own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically — which is also the intuition for why the explicit own-history blends fail: the state vector already carries the name's own recent information, so bolting it on again adds only noise. Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name's *own recent dynamics* (realized vol above all; removing it costs +1.0%, removing all cross-sectional characteristics only +0.5%), with characteristics mattering at cold-start. Two explicit hierarchical corrections — a naive quantile blend toward own-history and an amortized-as-prior affine Gibbs update — both *hurt* at every listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly. Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 on the M5 benchmark — benchmark tier — outside finance altogether.

Practitioner's selection rule. The two estimators minimize the same loss over different index sets — the trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\widehat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level; the network solves $\min_\phi \sum_i \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum — and the choice follows from what the application consumes. If the desk needs *numbers at fixed regulatory levels* (VaR/ES reporting, limits, capital), use the trees: best point accuracy, no GPU, no sampling needed. If the application consumes *draws* — scenario generation, portfolio aggregation by simulation — use the network: it is the only member that is an exact sampler, at a measured cost of ∼0.75% in point accuracy. If both are needed, run both from the same pooled residual panel; they share every upstream and downstream stage of (5), so the swap is one component, not a new pipeline.

Algorithm 2 (amortized shape model). (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names j and days t. (ii) Train H_ϕ by SGD on $\mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z} - H_\phi(s, \tau))$ [network], or boost per-τ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset — including one with no return history, using characteristics-only s — evaluate $H_\phi(s_t, \cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0, 1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

3.4 Formal results: why shape misspecification must show up in the loss

The paper's organizing claim — that the nonparametric edge lives exactly where the fixed innovation shape is locally wrong — is more than an empirical pattern: it follows from the structure of the pinball loss. The results below are elementary but do real work: they convert "the model's shape assumption is violated" into a quantitative lower bound on forecastable loss.

Lemma 1 (pinball regret identity)**.** *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ (so S is finite) and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F\, \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q. Then for any q,*

$$S(q) - S(q_F) \;=\; \int_{q_F}^{q} \big(F(u) - \tau\big)\, du \;\geq\; 0, \tag{7}$$

with equality iff $F \equiv \tau$ on the interval between q_F and q. If F has density bounded as $0 < m \leq f \leq M$ on that interval,

$$\tfrac{m}{2}\,(q - q_F)^2 \;\leq\; S(q) - S(q_F) \;\leq\; \tfrac{M}{2}\,(q - q_F)^2. \tag{8}$$

Proof sketch. The difference quotients of $q \mapsto \rho_\tau(Z - q)$ are dominated by the Lipschitz constant $\max(\tau, 1 - \tau)$, so differentiation under the expectation is justified and $S'(q) = F(q^-) - \tau$ (a.e.

equal to $F(q) - \tau$); integrating from q_F to q gives (7) *for any* F, atoms forming a null set. The integrand $F(u) - \tau$ has the sign of $u - q_F$ (by definition of q_F), so the integral is ≥ 0 whichever side q lies; (8) then follows from $F(q_F) = \tau$ (supplied by the density hypothesis) and $m|u - q_F| \leq F(u) - \tau \leq M|u - q_F|$. Full details, including both orientations of the integral, are in Appendix A. $\square$

Lemma 1 says the extra loss a misspecified forecaster pays is *quadratic in its quantile error*. The frontier claim then reduces to: does shape misspecification create a quantile wedge at the risk levels that matter? For the paper's canonical case — a fixed Student-t shape when the true local shape is fatter — it must:

Proposition 2 (tail wedge under kurtosis misspecification). *Let $Q_\nu(\tau)$ denote the τ-quantile of the unit-variance Student-t with $\nu > 2$ degrees of freedom. There exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2)$ such that for all $\tau < \tau^*$ and $\nu_1 < \nu_2$, $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$: the fatter-tailed law has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ at every level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof sketch. The unit-variance t_ν ($\nu > 2$, so the variance is finite and the normalization is well defined) has a regularly varying left tail: $\Pr(Z < -x) = c_\nu \, x^{-\nu}(1 + o(1))$ as $x \to \infty$ with $c_\nu > 0$; unit-variance rescaling alters c_ν but not the index ν. Inverting a regularly varying tail (Embrechts et al., 1997) gives $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$. For $\nu_1 < \nu_2$ the exponent $1/\nu_1 > 1/\nu_2$ makes $|Q_{\nu_1}(\tau)|/|Q_{\nu_2}(\tau)| \to \infty$, so the strict ordering and the diverging wedge hold for all τ below some $\delta > 0$; taking $\tau^* = \min(\delta, 1/2)$ places τ^* in $(0, 1/2)$ as claimed. (The proposition asserts *existence* of such a τ^*, not the location of the finite-sample crossover; the latter is $\tau_c \approx 0.0179$ for the t_3-vs-normal case of Remark 1.) The regret bound is (8); full details in Appendix A. $\square$

Remark 1 (the crossover is real and matters). Because unit-variance fat-tailed laws put *more* mass near the center as well as in the far tail, the quantile ordering of Proposition 2 reverses at moderate levels: numerically, the unit-variance t_3 has a *less* extreme 5% quantile than the normal (-1.36 vs -1.64) but a more extreme 1% quantile (-2.62 vs -2.33). The sign flip occurs at a single crossover level $\tau_c \approx 0.0179$ (solving $Q_{t_3}(\tau) = \Phi^{-1}(\tau)$ numerically for the lower-tail root), so the deepest deployable level $\tau = 0.01$ sits *below* the crossover (fat tail strictly more extreme) while $\tau = 0.025$ sits just *above* it (at 2.5% the unit-variance t_3 quantile is -1.837, still less extreme than the normal's -1.960). Two of the paper's design choices are direct consequences. First, this is why CRPS-style whole-distribution scoring can rank a fat-tail-aware model *below* a thin-tailed one even when the tail is the decision-relevant region (Section 3.1): body and far tail disagree in sign. Second, it is why the deepest deployable level straddles the crossover rather than sitting cleanly beyond it, and why the score of Section 3.7 is used as a *rank* rather than a moment estimate.

Lemma 2 (validity of the generative sampler). *If $\tau \mapsto \widetilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0,1)$, then $Z^* = \widetilde{q}(\tau \mid s)$ has quantile function exactly $\widetilde{q}(\cdot \mid s)$.*

Proof sketch. For any x, left-continuity and monotonicity make $\{t \in (0,1) : \widetilde{q}(t \mid s) \leq x\}$ an interval closed at its top, so $\{\widetilde{q}(\tau \mid s) \leq x\} = \{\tau \leq G(x)\}$ with $G(x) = \sup\{t : \widetilde{q}(t \mid s) \leq x\}$. Since $\tau \sim U(0,1)$, $\Pr(Z^* \leq x) = \Pr(\tau \leq G(x)) = G(x)$, i.e. Z^* has CDF G. As G is the generalized inverse of $\widetilde{q}$ and $\widetilde{q}$ is nondecreasing and left-continuous, the generalized inverse of G returns $\widetilde{q}$ exactly (the standard inverse-of-inverse duality), so $\widetilde{q}(\cdot \mid s)$ is the quantile function of Z^*. Full statement in Appendix A. □

Remark 2 (nesting and ERM dominance). The hypothesis class of Stage 2 contains the constant-in-s map $q(\tau)$, i.e., FHS; empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Section 6 test whether that dominance survives estimation noise. It does; the converse is that the same argument does *not* apply to CAViaR, which is why CAViaR is the one rival the engine only ties.

3.5 A synthetic boundary check: when the frontier does *not* appear

Before the real data, a simulation with known truth — run once, seed fixed — checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the thermometer: unit-variance Student-t residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). Three results. First, the Lemma 1 identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals (3.5×10^{-5} at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 promises. Third — the instructive one — the score-binned nonparametric forecaster does *not* beat the fixed shape here: a single $t_{\widehat{\nu}}$ fitted on the mixture ($\widehat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small — static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a t_ν law has infinite estimator variance for $\nu \leq 8$, so a moment-based thermometer is at its noisiest just when tails are fat, and sixty-three observations cannot reliably rank regimes that differ only in a static ν. For the empirical sections this means the real-data top-decile edge of Section 5 cannot be an artifact of stationary fat tails being averaged over — that configuration is the one the simulation shows a fixed shape handles — but must reflect *local* shape breaks a single fitted ν cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 negative control reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

3.6 What is different from Polson–Sokolov GBC

Because Polson and Sokolov (2023) is this paper's methodological benchmark, we state the deltas explicitly; each is an innovation forced by the downside-risk target, and each is defined mathematically above. (1) *Base measure*: GBC trains with $\tau \sim U(0,1)$; we train with the tail-weighted mixture $\lambda = \frac{1}{2}U(0,1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$, because uniform sampling starves the extreme levels that downside risk is about (the uniform-trained net's 99% breach rate was 3.2%). (2) *Input space*: GBC maps raw data to posteriors; we first pass returns through the parametric filter (2) and model *standardized residuals* — the scale dynamics are the one thing the industry model already gets right. (3) *Beyond-data tail*: no generative regression can know about quantiles beyond its training support; we splice the GPD tail (4) at p_0, replacing extrapolation by the network with extrapolation by extreme-value theory. (4) *Finite-sample guarantee*: GBC's output carries no coverage guarantee; the conformal stage adds an exchangeability-based one (Proposition 1), whose deployment status — motivation under time ordering, with the exception tests as arbiter — Section 3.2 states. (5) *Amortization target*: GBC amortizes across *simulations of one model*; we amortize across the real cross-section of hundreds of assets, which is what creates the cold-start capability and the transfer results of Section 3.3. (6) *Scope discipline*: GBC is a computational method; this paper adds the empirical question GBC does not ask — *when* such machinery beats the parametric filter at all — answered by the score below.

3.7 The misspecification score

GARCH-t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student-t with fixed shape. Static heavy tails do not violate this — ν absorbs them. What violates it is the shape being *locally* wrong: recent residuals far fatter-tailed or more asymmetric than any fixed t. We therefore score, per asset-day, over the trailing window $W_t = \{t-63, \ldots, t-1\}$ of $n_w = 63$ residuals—the window *ends the day before the forecast target*, so every signal is lagged at least one day relative to the loss it is paired with, and the frontier sorts of Section 5 pair S_{t-1} with the day-t loss (this alignment is enforced in the estimation code by construction) with window mean $\bar{z}$ and standard deviation s_z:

$$\text{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \qquad \text{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^3, \tag{9}$$

Terminology is fixed here once: *skewness* is the signed third moment sk_{63}; *asymmetry* always means its absolute value $|\text{sk}_{63}|$ (departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign); everywhere the text says "asymmetry," the statistic computed is $|\text{sk}_{63}|$. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-4, \ldots, t\}} |\widehat{z}_i|$. The frontier score is the decile rank of these within the trailing cross-section. All three are causal (trailing-window) and computable on any asset with a fitted GARCH — which is what makes the frontier a deployable monitor rather than an ex-post classification. Informally we will call this the misspecification *thermometer*: it reads the local temperature of the parametric assumption, and Table 2 is the treatment chart.

Why these statistics. The parametric model's one fragile assumption is that the standardized residual's *shape* is a fixed t_ν; its first two conditional moments are already absorbed by μ_t, σ_t. The lowest-order ways a fixed symmetric law can be locally wrong are precisely the third and fourth standardized moments: sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies (an exact t_ν sample has known positive excess kurtosis; the score is used as a *rank*, so only its cross-sectional and temporal variation matters, not its t_ν baseline). J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent — and Section 5 shows the realized edge concentrates there. Note the logical status of this choice: after μ_t and σ_t are fitted, the fixed innovation shape is the parametric model's *only* remaining free assumption. A meter that measures the local violation of that one assumption is therefore not one candidate signal among many — it is the residual degree of freedom itself, which is why one should expect it, a priori, to be the axis along which flexible-shape models win.

Algorithm 4 (the misspecification meter, as deployed). (i) Fit (2) on a trailing window; store $\widehat{z}_{t-62}, \dots, \widehat{z}_t$. (ii) Compute mk_{63}, $|\mathrm{sk}_{63}|$, J_5 by (9). (iii) Convert each to a decile rank against the trailing pooled panel (same asset class); the score is the rank of the maximum. (iv) Act by Table 2: top decile $\Rightarrow$ the standard parametric VaR is currently, locally wrong by $\sim$2–3% of pinball with high confidence — flag the asset for the nonparametric / residual-hybrid engine and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter serves surveillance, and day-level switching is unsupported, Section 5.3). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

4 Data and evaluation protocol

Equity panels come from CRSP daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations; every modeling choice in the paper was made on this panel. The confirmatory holdout covers 2000–2013 and was pulled once, after the specification was frozen: common shares (share codes 10 and 11) with at least 3,000 observations in the window, ranked by average market capitalization, top 200. Returns are in percent and include CRSP delisting returns where present; the one delisting event in the design panel moves headline results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data. Terminal access ended in mid-2026, so those exhibits are frozen at the version reported here and all continuing work runs on WRDS alone.

Licensed rows never leave the machines they were pulled to. The public repository carries code, configuration, and derived statistics; the panels rebuild from the queries above for any licensed subscriber. Table 3 gives the sample flow.

Splits are chronological within each name. The first 60% of a name's history is the estimation window and the last 40% is test; the final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH-t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on pooled estimation-window residuals, the GPD tail on the same pooled residuals below their empirical 2.5% point, and the conformal shifts on the pooled calibration split. No stage refits inside the test window unless a rolling variant is named explicitly. Two structural objections to this protocol — calendar overlap in pooled training, and the universe rule's survivorship — get their own subsections at the end of this section, each with the test that settles it.

Loss and test conventions are collected in the conventions note that opens Section 5: mean pinball loss across eleven quantile levels for distributional accuracy, the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019) for joint (VaR, ES) pairs, per-date Diebold–Mariano statistics with a Newey–West variance for loss comparisons, the Model Confidence Set for multi-model fields, and Kupiec, Christoffersen, and date-clustered breach tests at the two regulatory levels for exception behavior.

"Registered" has a specific meaning in this paper. A run is registered when the full specification — signals, features, hyperparameters, split rule, and test statistics — is frozen and the predictions are written down before the data are pulled. The holdout pipeline is line-for-line the design-era pipeline — both files are public in the replication repository — so a reader can diff the two and verify that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 5 ran under this rule. Registered runs that remain open are listed in Section 8.

4.1 Common shocks, calendar overlap, and the pooled learner

The design panels are split per name, 60/40 in each name's own time index. In an unbalanced panel such a rule can place one name's training window on the calendar dates of another name's test window, and a learner pooled across names could then meet a systemic event — a common volatility shock, a market-wide liquidity break — during training that a per-name model would only ever meet out of sample. The channel requires no individual feature to look ahead; the leak, if present, rides the cross-section through calendar time. We concede the point in principle: a per-asset split cannot by itself support a causal reading of a pooled model's edge. In these particular panels the channel is nearly degenerate — both universes select names with near-complete histories, so per-name split points fall on almost the same calendar date — but nearly is not zero, and the question deserves an answer by construction rather than by argument.

The construction is a strict calendar-time split: every stage — each GARCH fit, the pooled residual learner, the GPD tail — is re-estimated on data strictly before 2020-01-01, and testing starts at that date for every name simultaneously, so no training window of any name shares a single calendar day with any test window of any name. The frontier survives. The top misspecification decile carries a +2.47% edge with per-date DM 6.22 over the 1,258 test dates (compare +2.7%, DM 6.5, under the per-name split); the overall edge is +0.71% (DM 1.87). The profile's bottom decile is elevated in this split (+1.3%: a 2020–22 test era leaves few truly calm cells), but the ordering that matters is intact — the top decile roughly doubles every other cell. The pooled edge, and its concentration where the score is high, is not a calendar-overlap artifact. A second cut targets the harder era: inside the 2000–2013 holdout, everything is re-estimated strictly before 2007-01-01 and tested on 2007–2013, so the global financial crisis is entirely absent from training for every name. The pooled learner still wins overall (+0.29%, per-date DM 3.08 over 1,762 dates) and the within-sample decile profile keeps its shape (top three deciles +0.45–+1.04% against +0.06–+0.09% in the bottom three); the frozen-threshold top cell of Section 5 is too thin in that era to test alone (3.4% occupancy, +0.85%, DM 0.7). A model that never saw a systemic crisis in training carrying a significant edge through the crisis it never saw is the opposite of what the leakage story predicts.

4.2 Cross-sectional dependence, family-wise error, and the universe rule

Three inference-level objections deserve the same treatment. *Dependence:* all headline tests already collapse the cross-section to one number per date before any variance is estimated (Section 5, conventions), so common-date dependence enters the test through the date series itself rather than being ignored; block-bootstrap procedures (SPA, MCS) resample dates, not asset-days. *Multiplicity:* the frontier grid spans three signals and ten deciles, and per-cell p-values ignore the thirty hypotheses' joint draw. We therefore report Romano–Wolf step-down family-wise-error control over the full 3×10 grid: per-date mean edges per cell, a stationary bootstrap over dates (mean block 10 days, $B = 2{,}000$; block 20 as sensitivity) that preserves cross-hypothesis dependence, and one-sided step-down adjusted p-values. Nine of the thirty cells survive 5% family-wise control under both block lengths, and they are the frontier's own: the top deciles of all three signals (mk_{63} $t = 9.4$, skew_{63} $t = 9.2$, jump_5 $t = 5.1$, adjusted $p < 0.001$), the ninth deciles of mk_{63} (adjusted $p = 0.022$) and skew_{63}, skew_{63}'s seventh, and jump_5's U-shaped far cells. Every mid-grid cell with an unadjusted t near two — mk_{63}'s sixth decile ($t = 2.34$), jump_5's seventh ($t = 2.08$), skew_{63}'s first ($t = 2.01$) — loses significance once the family is priced, and the text treats them accordingly: the paper's claims rest on the cells that survive. Any cell that loses significance under the adjustment is identified as statistically indistinguishable from zero in the text wherever it is discussed. *Survivorship:* the holdout's observation filter conditions on names present for most of the era; the frontier comparison is relative — both models score the same names — so the filter tilts the universe toward survivors without by itself favoring either model in the ranking. The sharper version of the objection is selection on information unavailable at the start, and it too has a constructive answer: a point-in-time re-selection fixes the universe at the top 200 names by market capitalization as of the first quarter of 2000, applies no minimum-history filter, and keeps every name through its delisting date with the CRSP delisting return appended as the final observation. The point-in-time run is the strongest replication in the paper: on the 164 of 200 Q1-2000 names that clear the symmetric fitting floors (36 dropped; 79 delisting returns ridden to the end), the overall edge is +0.58% with per-date DM 8.96 over 2,692 dates, and the frozen design-era top bucket carries +2.94% at DM 6.6 (5.9% occupancy; the bottom bucket is +0.44%, DM 5.4 — in this era even calm cells lean positive, at one-seventh the top bucket's size). The edge is *larger* than in the survivor-tilted holdout: the observation filter turns out to have biased the reported edge downward, not upward. Names that later die are where shapes break; a universe that keeps them is kinder to the flexible engine and harder on the leakage-and-selection story.

5 The misspecification frontier

Conventions. Conventions are fixed once, here. (i) *Edges* are percentage reductions in pinball loss, so *larger is better* for the named model; a "+2.7% edge" means the model's average loss is 2.7% lower than the benchmark's. (ii) *Ratios* are model-over-benchmark loss ratios, so *below 1 favors the model* and $1 - \text{ratio}$ is the edge (0.973 = a 2.7% win; 1.043 = a 4.3% loss). (iii) *DM* is the Diebold–Mariano t-statistic on the loss difference. *Per-date DM* means, concretely: losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$, and the test is run on the resulting $\sim$1,100-date series with a Newey–West (lag-10) variance—the effective sample is the number of dates, never the number of asset-days. Reported p-values are *one-sided* in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%); t-statistics above 2.6 are significant under either convention, and marginal cases are flagged where they occur. A *tie* is a bounded claim, not an absence of evidence: the label is used only where the per-date DM cannot reject zero at the two-sided 10% level *and* the point edge lies inside ±0.25% of the benchmark's loss — an explicit equivalence margin. The Model Confidence Set is computed on the same per-date loss matrix with the range statistic and a stationary bootstrap (mean block length 10 days; $B = 1{,}000$ replications in the main battery, $B = 800$ in the CAViaR extension), following Hansen et al. (2011). (iv) *Coverage* numbers are realized frequencies against a stated nominal (0.05 target: 0.056 is slight under-coverage of risk, 0.032 over-conservatism); for *credible-interval* coverage, closer to nominal is better in both directions. (v) *Correlations* quoted for the frontier relate a universe's residual-kurtosis level to the size of the nonparametric edge across that universe's members. No other significance marks are used.

5.1 The within-universe result

On 200 CRSP names (≈221.6k test asset-days), we bucket the per-day pinball edge of the amortized nonparametric model over GARCH-t into deciles of each misspecification signal. Table 4 shows the result that organizes the paper.

The edge is a top-decile phenomenon. It is large and significant where the fixed-t shape assumption is currently broken, and statistically indistinguishable from zero where it is not.

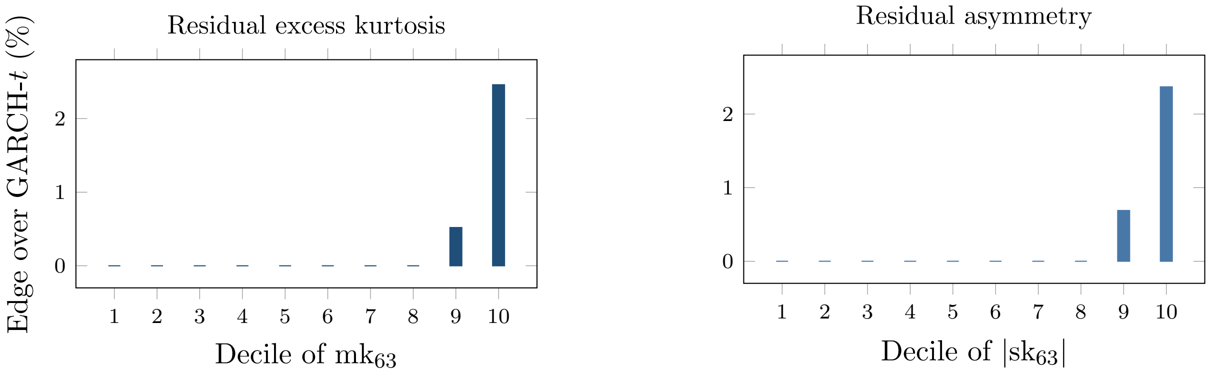

Figure 2: The misspecification frontier. Nonparametric pinball edge over GARCH-t by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are ≈ 0 (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.46% kurtosis, +2.37% asymmetry; per-date DM 6.5 in the top kurtosis decile).

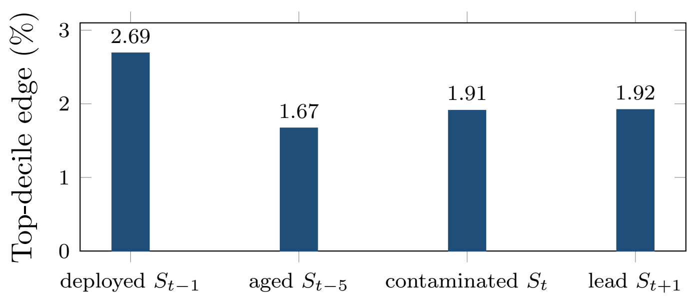

Figure 3: Timing diagnostics for the frontier. Top-decile edge under four alignments of the mk_{63} score with the daily loss (200 CRSP names; per-date DM 5.5–6.5 in all four). The deployed lag dominates, and the contaminated same-day alignment yields less — the opposite of what a look-ahead artifact would produce.

Timing diagnostics. The score is computed from residuals through day $t-1$ and paired with the day-t loss; Figure 3 probes that alignment. The deployed lag gives the largest top-decile edge (+2.69%, DM 6.5). A five-day-old score retains +1.67% (DM 6.4), so the signal is persistent enough to act on. A deliberately contaminated same-day window — the alignment a look-ahead artifact would favor — yields *less* than the deployed lag (+1.91%), and a one-day-ahead placebo matches the contaminated value rather than exceeding the deployed one (+1.92%; with a 63-day window the score's one-day autocorrelation is near one, so lead placebos are weak by construction and the informative comparison is deployed-vs-contaminated). Same-day sorting does not manufacture the edge.

An absorption test. If the frontier score were merely an omitted variable, a parametric-adjacent fix should capture the edge without any pooled learner: run filtered historical simulation per name, but reweight the historical residual window toward days whose mk_{63} resembles today's (Gaussian kernel, bandwidth half the signal's per-name standard deviation). We ran that absorber on the full panel. The reweighted FHS is significantly *worse* than plain FHS overall (per-date DM -11.3) and much worse in the top misspecification decile (average pinball 0.541 against 0.444, DM -11.1), while the pooled shape model beats plain FHS in that decile by 2.7% of pinball (DM 8.8) and GARCH-t by 2.8% (DM 9.1). The mechanism is effective-sample collapse: sharp univariate conditioning shrinks a name's usable history to the handful of days that resemble today, exactly when today is unusual; smoothing the kernel only interpolates back toward the unconditional FHS it was meant to improve. Pooling across names is what makes state-conditioning affordable — the amortized model borrows the shape information from every name's stressed days. The amortization of Section 3.3 is a precondition for the frontier result; a single name's history cannot support sharp state-conditioning on its own.

Robustness of the score. Table 5 double-sorts the edge on the score and on trailing volatility. Sorting test asset-days independently into quintiles of mk_{63} and of trailing volatility (rv_{63}): high volatility without a shape break carries no edge (top-volatility column, bottom four kurtosis quintiles: -0.57% to $+0.10\%$), a shape break without an active regime carries almost none (top-kurtosis row, calmest quintile: -0.10%, DM 0.7), and the product state carries $+2.33\%$ with DM 6.4. The two ingredients interact. The score flags a currently wrong parametric shape; volatility scales the cost of the error. Second, a superior-predictive-ability test (Hansen, 2005) on the per-date loss panel against the full desk field (GARCH-t, FHS, EWMA, HS) returns $T^{\mathrm{SPA}} = 0$, consistent $p = 1.00$: no rival's advantage is even positive, and each is individually worse with $t \geq 5.8$. Third, the score needs no recalibration to stay useful: holding the design-era top-decile threshold fixed, the top-decile edge is positive in every test year (between $+1.7\%$ and $+3.4\%$, 2020–2024), even though the cross-sectional 90th percentile of mk_{63} itself drifts upward in 2023–24. Fourth, the score is a slow state, so monitoring it costs little: a median name enters the top decile less than once a year (0.68 entries) and a median episode lasts 24 trading days.

The confirmatory holdout. Every design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze the entire pipeline — signals, features, hyperparameters, split rule, and test statistics — registered three predictions (top-decile edge positive and significant; bottom decile indistinguishable from zero; a threshold-shaped decile profile), and then pulled CRSP daily returns for 2000–2013, an era no prior run in this project had touched. The frontier replicates (Figure 4): the top-decile edge is $+0.89\%$ with per-date DM 2.16 ($p = 0.016$; 1,465 dates), deciles one through nine sit between -0.18% and $+0.28\%$, and the overall edge is $+0.02\%$ — the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. The holdout spans the 2008 crisis, so the replication is not an artifact of a calm era. One measurement choice in the replication deserves its own audit: decile boundaries above are within-sample ranks of the holdout test era (the standard conditional-sort convention), so the partition rule — though it never touches a forecast or a loss — is not a rule a desk could have applied on day one. The restatement a desk could have run: freeze the design era's numeric decile boundaries (the pooled design test sample's mk_{63} edges, top threshold 13.67) and apply them to the holdout as constants, so membership at (i, t) depends only on name i's own past 63 days and a number fixed before any holdout day was scored. The threshold shape survives

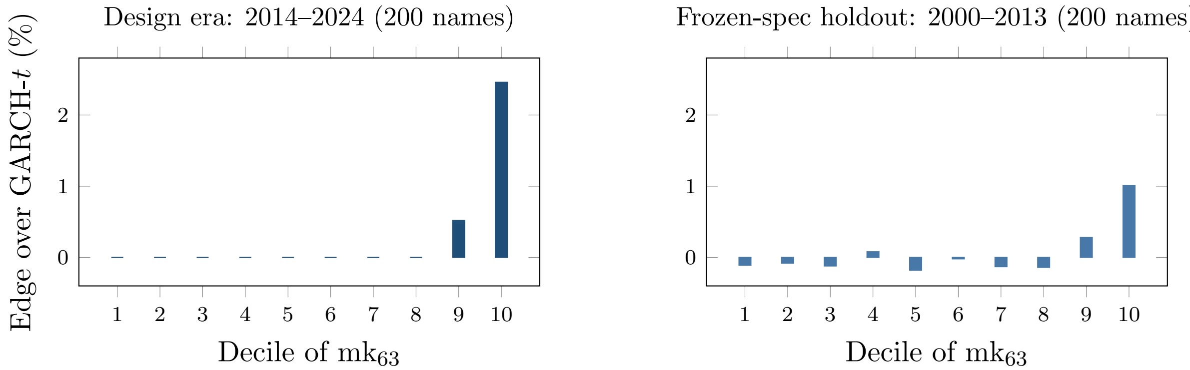

Figure 4: The frontier out of era. Left: the design-era decile profile (Table 4). Right: the same pipeline, frozen and run once on 2000–2013 CRSP data under registered predictions. Bars are pooled per-observation decile means (top decile +1.01%); the inference statistic is the per-date aggregation of Section 3.7, +0.89% with DM 2.16 over 1,465 dates. All other deciles sit between −0.18% and +0.28%. The threshold shape survives an untouched era containing the 2008 crisis.

the frozen rule — bins eight through ten carry +0.82%, +0.92%, +1.02% against −0.3–0.0% below — and the top-bin point edge is slightly *larger* than the within-sample sort's (+1.02% vs +0.89%), but the cell thins to 3.3% occupancy in this lower-kurtosis era and the per-date test loses power (DM 1.11, insensitive to Newey–West lags 5–44). The frozen rule's decisive test is the point-in-time universe of Section 4.2, where the same fixed thresholds meet a universe selected with no later information and deliver +2.94% at DM 6.6.

The low overall correlation between signal and edge (0.05–0.10) is not a weakness of the story but its point: the relation is a threshold nonlinearity, not a linear trend, and the top-decile Diebold–Mariano statistics are the correct summary.

5.2 The frontier across universes

Table 6 shows the same axis at work across markets. Three features stand out. First, the only decisive *standalone* single-asset wins anywhere are the hyperinflation and capital-control currencies — Argentine and Egyptian pesos, 12–14% better with significance — whose residual kurtosis (∼2000–3300) sits two orders of magnitude above anything GARCH-t was designed to absorb. Second, the country gradient is mediated by residual kurtosis, not by the development label per se: frontier indices (kurt ∼10) sit between well-behaved developed indices (kurt ∼5) and hyperinflation FX on the same frontier. Third, the Korea-2026 negative control sharpens the mechanism: a dramatic *price* crash with circuit breakers is *not* residual misspecification — Korean residual kurtosis stays in the ordinary 4–13 range, and GARCH-t significantly wins all twenty-three assets even inside the crisis window. The frontier is about the residual shape, not about headline turbulence.

A frequency corollary. Daily crypto belongs to GARCH (zero nonparametric wins on five liquid coins). At *hourly* frequency the picture inverts: the neural IQN beats the full benchmark

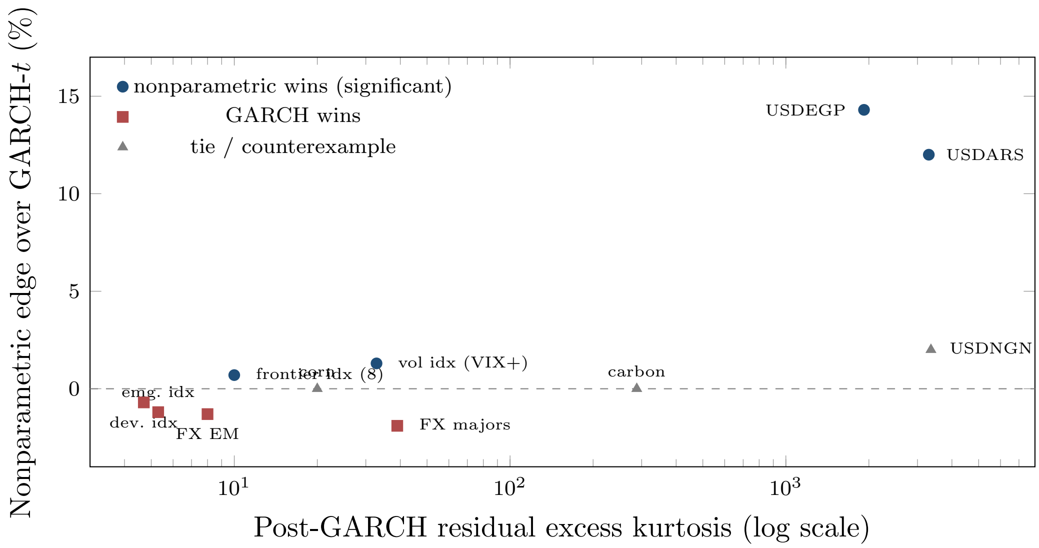

Figure 5: The frontier across universes: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis ∼2000–3300) carries the only decisive standalone wins; high kurtosis is necessary but not sufficient — carbon (288) and corn (20) tie, and NGN's win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

ladder, including filtered historical simulation, by ∼0.6% with per-date DM $p < 10^{-7}$. Sample density is itself a frontier dimension: more observations per unit of regime raise the resolvable residual-shape signal. A systematic intraday map of the frontier is registered and blocked only on tick-data access.

5.3 Regime concentration and the futility of gating

In the amortized panel the edge concentrates in turbulence: the GARCH-minus-GBM pinball gap grows monotonically from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent — roughly fourfold — while both nonparametric estimators beat GARCH in *every* quintile (overall: GARCH 0.6543, GBM 0.6447, IQN 0.6478). A natural inference is that one should *gate*: run GARCH on calm days and switch to the nonparametric model when the score is high. It fails, instructively. A classifier gate trained on the misspecification signals routes 90.5% of days to the nonparametric model and exactly matches always-nonparametric (0.3987 vs GARCH 0.4000); worse, its routing runs *backwards* across score deciles, because nonparametric wins on high-score days are rare-but-large while a classifier targets frequency. A regression gate on the edge magnitude fixes the direction (routing flat ≈ 1.0, predicted-edge correlation +0.16) and still converges to always-nonparametric. The per-day oracle gap (4.4%) is unreachable because per-day model advantage is dominated by *which day the tail event lands* — unpredictable from prior-day state. The deployable conclusions: in the amortized setting, use the nonparametric model always (it does not underperform on calm days); use the residual-hybrid where a guaranteed ≥GARCH floor is wanted; and use the frontier score as a *monitor*

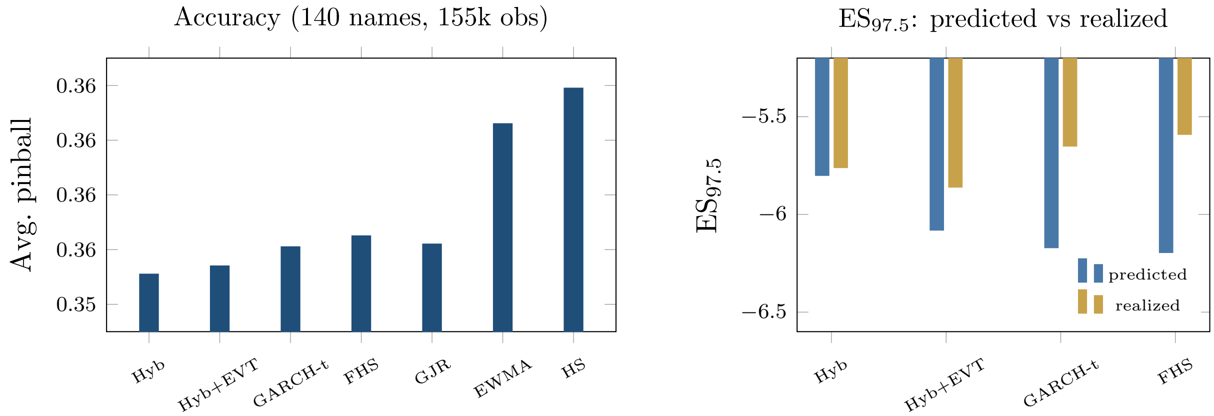

Figure 6: The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 4.4–8.3 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: $\mathrm{ES}_{97.5}$ predicted vs realized — FHS and GARCH-t over-state ES by 8–11% (over-charging capital) while the hybrid is nearly exact.

(Section 7); day-level switching adds nothing. (The score's detection latency is small — median zero, mean two to four days — and because the nonparametric edge is back-loaded within misspecification episodes, the lag costs little even in the per-name setting, where gating does matter because standalone nonparametric models underperform on calm days.)

6 Beating the industry standard: the FRTB battery

6.1 Head-to-head accuracy and significance

Table 7 reports the full battery on 140 CRSP names (155k obs, 1108 dates).

The stress-era replication. The battery's registered re-run on the 2000–2013 holdout panel is complete. Splits follow Section 4, so each name's test window runs from roughly mid-2008 through 2013 — the crisis and its aftermath. The EVT-tailed engine passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t-statistics of 0.17 at 99% and 0.35 at 97.5% — consistent with nominal at both regulatory levels through the crisis window. The two models most common on desks do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH-t is also well calibrated in this era (94% pass rate), which is the frontier's own prediction — calibration parity is the norm, and the engine's added value in any era is the accuracy edge where the misspecification score is high. The conformal layer, calibrated on pre-crisis data, tilts conservative out of era (breach 2.21% at 97.5%, $t = -1.09$, not significant): the safety margin widens when the calibration era is calmer than the test era, and the direction of the error is the one regulators prefer.

Three caveats qualify the headline. First, adding SAV-CAViaR — absent from banks' standard toolkit but the strongest semiparametric academic benchmark — produces a statistical tie:

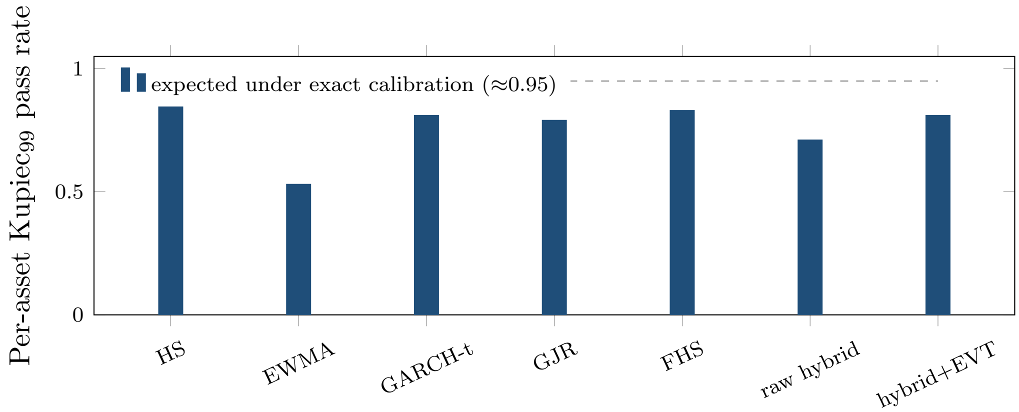

Figure 7: Exception tests restated per asset (140 CRSP names, 99% VaR): share of names whose individual Kupiec test is consistent with nominal. HS over-covers (conservative); EWMA fails; the EVT-tailed engine matches the parametric benchmarks while beating them on loss and ES calibration (the GJR bar predates the skew-t correction of Table 7's note; the corrected pooled row passes both 99% tests) (Table 7). Date-clustered tests in the table notes.

CAViaR 0.3461 vs hybrid 0.3460 (DM 0.59, $p = 0.28$; the CAViaR study's common sample differs slightly from Table 7's, which is why the hybrid's loss level is 0.3460 there against 0.3551 in the table — rankings, not levels, are comparable across the two runs); the 90% Model Confidence Set contains exactly these two. The hybrid is *co-best with CAViaR*, not uniquely dominant, and still significantly beats everything banks actually run. Second, the neural IQN as residual model initially loses to trees badly (0.3650 vs 0.3551, DM 13.3) with a severely thin tail (breach_{99} 3.2%); the tail-aware, monotone, conformally recalibrated version closes to a hair (0.3638 vs 0.3632, DM 3.03) with calibration repaired (breach_{99} 0.92%). Trees remain the strongest tabular estimator; the tuned IQN is competitive. Third, $\text{ES}_{97.5}$ calibration — the actual capital metric — is where the hybrid's advantage is most consequential: FHS and GARCH-t both *over-state* ES by 8–11%, i.e. over-charge capital, while the hybrid is nearly exact.

The joint (VaR, ES) **score.** Because ES is not elicitable alone but the pair (VaR, ES) is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the battery was re-checked under the FZ0 joint score (orientation validated by Monte Carlo against a known-truth t_5 before any ranking was read; first sandbox subsets of the panel, full-panel re-run registered). Three results. On a 45-name large-cap subset, a Gaussian-innovation GARCH is significantly *worst* at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, fat tails are strictly necessary. Among the fat-tailed entrants — GARCH-t, FHS, and the hybrid+EVT — FZ0 differences on large caps are statistically indistinguishable ($|\text{DM}| < 1.4$): on the well-behaved segment the EVT stage's distinct value is its exception-test compliance, a calibration property the FZ0 score does not reward. On a 40-name top-kurtosis subset the ordering flips and the hybrid+EVT attains the lowest FZ0 — the frontier pattern reappearing under a jointly consistent score — directionally but not yet significantly at this width (DM ≈ 0.9, $p \approx 0.17$).

The registered full-panel re-run decided it. On the full 200-name panel (221.6k test asset-days), with ES from the fitted GPD tail and per-date DM inference throughout, the engine's *accuracy layer* — the EVT-tailed forecaster with no conformal shift — attains the lowest FZ0 at both regulatory levels: at 1% it beats GARCH-t (DM 4.1), FHS (DM 4.7), and a per-name score-driven one-factor GAS model estimated by FZ0 minimization in the style of Patton et al. (2019) (DM 9.2, zero fit failures); at 2.5% it again wins (DM 4.2 over GARCH-t, 6.0 over GAS). The frontier concentrates the joint-score edge as it does the pinball edge — the top misspecification decile carries DM 2.4 at the 1% level against DM 0.2 in the panel bulk. Adding the conformal shift at 97.5%, whose calibration split includes the 2020 crash, widens the band out of era (breach 1.6% against the 2.5% target) and gives back the FZ0 advantage there: the *deployed* variant loses to GARCH-t on the 2.5% joint score (DM -2.0, marginal at the two-sided 5% level), and the concession is largest exactly where the bands are widest — in the top misspecification decile the deployed shift's joint-score deficit reaches DM -7.1. The coverage property costs sharpness under a strictly consistent score, and its error direction is conservative — wider bands, fewer breaches than nominal. A desk that prizes the conservative coverage margin keeps the shift; one scored on FZ0 efficiency runs the EVT tail alone, which passes the exception battery on its own (Figure 7). Every headline in this paper that says "lowest FZ0 at both levels" refers to the accuracy layer; the deployed variant's 2.5% concession is the quantified price of its coverage margin, not an unreported loss.

6.2 The ten-day horizon and stress

At the FRTB ten-day liquidity horizon (CRSP 2014–2024, directly-learned ten-day quantiles vs GARCH $\sqrt{h}$-scaling) the pinball edge grows from $\sim$0.4% at $h = 1$ to $\sim$1.4% at $h = 10$, and holds in the 2020–2022 stress window (hybrid $1.247 <$ FHS $1.2605 <$ GARCH 1.262). Two horizon designs appear in this paper and their numbers are not comparable: the stress battery here scores the *deployed engine's* h-day quantiles against $\sqrt{h}$-scaled GARCH on overlapping cumulative returns (0.4%$\rightarrow$1.4%), while the separate direct-learning study of Figure 8 trains a fresh quantile model *at each horizon* and finds larger ratios (3.4% at h=1 to 4.9% at h=20) — the baselines and estimands differ, and each figure names its own.

More consequentially for capital: ten-day GARCH $\sqrt{h}$-scaled ES is badly over-stated (predicted -19.4 vs realized -15.4) while the directly-learned hybrid is calibrated (-19.05 vs -17.70) with the best ten-day breach$_{97.5}$ (0.0265 vs target 0.025; GARCH 0.033). Fat tails compound with horizon; $\sqrt{h}$ Gaussian-style scaling compounds the shape error. A caveat: the temporal split places the whole test window in 2020–2024, and the registered rerun on 2000–2013 shows the ten-day result is era-dependent. Out of era the two methods tie on accuracy (direct edge +0.65%, date-level DM 0.22, NW lag 20), and the calibration verdict reverses: $\sqrt{h}$-scaled GARCH-t ES is almost exactly calibrated through the crisis window (predicted -16.6 vs realized -16.6) while the direct-learned tail average under-states it (-14.5 vs -17.6) — the pooled ten-day tail, trained on the pre-crisis segment, did not extrapolate to 2008-scale ten-day losses. The ten-day capital advantage of Section 7 is therefore a 2014–2024 statement, and a desk holding

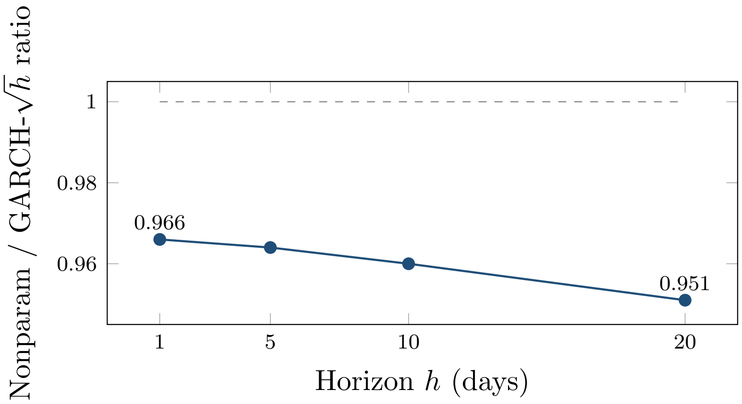

Figure 8: The edge grows with horizon. Nonparametric/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (amortized panel): fat tails compound while $\sqrt{h}$ scaling compounds the shape error — ∼5% at $h = 20$. Direct-learning design: a fresh quantile model per horizon against $\sqrt{h}$-scaled GARCH-t; the deployed engine's own ten-day numbers (0.4%→1.4%) are the stress battery's and are smaller — the designs differ and are labeled where quoted.

the engine at $h = 10$ through a 2008-type era should keep the parametric $\sqrt{h}$ number alongside as the conservative envelope. The frontier itself is confirmed on 2000–2013 (Section 5); this horizon result completes the registered comparison with a mixed verdict, reported as such.

6.3 Cross-asset winners, with corrections

The clean calibrated cross-asset win is VIX (pinball 1.127 vs 1.137, DM 1.74, $p = 0.041$ one-sided, passes Christoffersen at both levels)—a marginal result: it does not survive a two-sided convention, nor a multiplicity adjustment across the 43-instrument sweep, and is reported as suggestive rather than confirmatory. Credit (52% accuracy edge, DM 30.6) and Baltic freight (15%, DM 9.6) fail breach-independence — exceptions cluster — and are reported as accuracy edges with open calibration problems, not wins. (These are the FRTB-grade single-asset fits. Freight's magnitude agrees with the lean hybrid-vs-GARCH sweep of Table 6 (ratio 0.846 there); credit does not — the lean sweep shows only 4% (DM 2.3) — because the two batteries pair different model variants against different benchmarks. Both are reported; the batteries are not interchangeable, and credit's larger figure should be read with that caveat in addition to its clustered exceptions.) An earlier electricity "win" was an artifact: log-returns on near-zero and negative power prices; refit on price *differences*, GARCH wins ERCOT (DM −2.55) and PJM (DM −2.73), and electricity is removed from the win list. The GPD tail also *hurts* some volatility series (VVIX, V2X, MOVE flip from win to tie/loss), motivating the selective-application rule of Section 3.2.

7 Applications and deployment

The production loop. *Nightly:* (i) update each name's GARCH filter state $\widehat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector s_t and evaluate the shared shape model $q_\phi(\tau \mid s_t)$ — one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below p_0 and the conformal shifts c_τ; (iv) report $\widehat{Q}_{t+1}(\tau) = \widehat{\sigma}_{t+1}\, \widetilde{q}(\tau \mid s_t)$ at the desk's levels, with $\mathrm{ES}_{97.5}$ from the tail average; (v) read the misspecification score (9) as the model-risk monitor — top decile means the parametric fallback is currently ∼2–3% wrong on that name. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* evaluate (ii)–(iv) from characteristics and first ticks; no history required. Total marginal cost per name per day: one filter update and one model evaluation.

Ex ante use of the score. Because the frontier score is a trailing window on GARCH-standardized residuals, it is available *ex ante*: a desk computes today's score from today's data and acts on it for tomorrow's risk number. What the evidence does *not* support is day-level model-switching. As Section 5.3 shows, the per-day oracle gap is unreachable (predicted-edge correlation +0.16), because which day a tail event lands is not forecastable from prior-day state, and a learned gate collapses to always-nonparametric. The edge is, in effect, negligible over the calm ∼90% of asset-days and concentrated (+2.7% pinball, DM 6.5) in the top misspecification decile and the most turbulent volatility regimes (an ∼fourfold gap, Section 5.3). A desk should therefore run the robust engine always and read the score as a monitor: the nonparametric/residual-hybrid family *ties* the parametric benchmark on calm days and wins in stress, so a desk pays nothing to hold it and collects the stress-period edge automatically. The one setting in which the score does drive a switch is the per-name/standalone model, where nonparametric quantiles underperform on calm days; there the score gates, and its short detection latency (median zero, mean two to four days) costs little because the edge is back-loaded within a misspecification episode.

Regulatory capital. The hybrid+EVT+conformal engine is an internal-models VaR/ES candidate that tops the accuracy table while passing both exception tests at both levels. Under the internal-models approach the modelled charge is linear in $\mathrm{ES}_{97.5}$, so a model that over-states ES by a fraction δ_1 over-charges the modelled component by the same fraction, and replacing it with a model of over-statement δ_0 releases $(\delta_1 - \delta_0)/(1 + \delta_1)$ of that component. The measured calibration gaps of Section 6 then price out as follows, per 100 basis points of ES-driven capital:

Replacement	$\delta_1 \to \delta_0$	Release	bp per 100bp
FHS → engine, h=1	10.9% → 3.8%	6.4%	6.4
GARCH-t → engine, h=1	9.2% → 3.8%	4.9%	4.9
$\sqrt{h}$-scaled → direct, h=10	26.0% → 7.6%	14.6%	14.6

The ten-day row matters most because ten days is the FRTB base liquidity horizon, and it carries a caveat. The 26% over-statement is measured on 2014–2024, and the registered 2000–2013 rerun (Section 6) finds $\sqrt{h}$ almost exactly calibrated in that era — the ten-day release is a regime-dependent gain, realized in eras like the last decade and absent in a 2008-type era. The one-day rows are the era-robust component. The same fractions carry a certainty-equivalent reading: a desk sizing positions against a mean–ES mandate leaves exactly the released fraction of its risk budget idle when it runs the over-stated model, so the capital arithmetic and the maximum-expected-utility arithmetic coincide, and neither purchase comes with added breach risk (the engine's exception record is the battery's best). This is illustrative arithmetic on measured calibration gaps — the modelled component only, holding the stress-period calibration and non-modellable add-ons fixed — and not a capital plan.

A misspecification monitor. The frontier score is a live, per-asset monitor: when a desk's asset enters the top decile of residual-shape extremity, its standard VaR model is currently, locally wrong by ∼2–3% of pinball with high confidence. This is deployable as model-risk surveillance because it is real-time and cheap (a trailing window on GARCH residuals).

Cold-start risk for new listings. The amortized model is the only entrant in this paper that can produce a conditional risk forecast for an asset with *no return history*: an IPO, a spin-off, a newly listed coin or ETF gets day-one quantiles from its characteristics and its first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names. The practical numbers: the amortized forecast beats own-history benchmarks at *every* listing age with the edge largest when young (∼6–10% of pinball in the first month of trading, full-scale panel), and below roughly 250 trading days there is no fitted per-asset alternative at all — a GARCH cannot be estimated. The ablation of Section 3.3 tells a desk how the machine works as history accrues: characteristics carry the forecast at birth; the name's own recent realized volatility takes over as it ages, because the model has learned one universal vol-state-to-quantile mapping rather than per-name parameters. Risk limits, margining, and market-making inventory for the newest names are the immediate use cases.

Trading implications. The universes where the nonparametric tail wins — crisis FX, volatility indices, credit spreads, frontier equities — are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location: a correct physical-measure tail forecast does not by itself make bought crash insurance profitable, because the crash risk premium can survive conditioning on a correct forecast. The deployable side is the other one — *warehousing* (selling) the premium — with the model supplying tail control rather than alpha on the mean: model-timed overlays improve tail outcomes, not average returns, consistent with the tail being the only place the model knows something a fixed-shape filter does not. The contribution to a premium-harvesting book is risk management (when to size down or step aside ahead of a genuine crash), not signal generation.

Scenario generation. Because the shape stage carries an exact inverse-CDF sampler, calibrating intractable scenario generators (rough volatility, Hawkes, jump-diffusions) for CCAR/ICAAP-style stress design is a natural extension; it is not developed here.

8 Conclusion and limitations

A fixed-shape parametric filter is hard to beat where its assumption holds, which is most assets on most days. The paper's answer to "when and why beat it" is a measurable frontier: local violation of the residual-shape assumption, at whose top decile the distribution-free family wins decisively, significantly, and portably across four universes. The engine that captures this — parametric scale, nonparametric shape, EVT tail, conformal finish — meets and exceeds the FRTB standard banks actually run. The generative members of the family contribute further capabilities that no point model can — calibrated uncertainty on the risk number, and likelihood-free posteriors for models whose likelihoods do not exist in closed form — natural extensions beyond this paper's scope.

Limitations. Gradient-boosted trees remain the strongest tabular estimator; the neural IQN earns its place through sampling and amortized posterior inference rather than point accuracy. The frontier score is necessary-not-sufficient (carbon/corn counterexamples; Baltic Dry wins at moderate kurtosis via limit-move dynamics). Credit and freight show large accuracy edges but clustered exceptions — unresolved. The frontier's 2008 exposure is covered by the 2000–2013 holdout, and the FRTB battery's stress re-run on that era is complete (Section 6: the engine passes both levels through the crisis window). Intraday extension is blocked on tick-data access. And the per-day oracle gap of Section 5.3 should discipline the field's enthusiasm for model-switching: most of the day-level advantage of any model pair is realization noise, not predictable structure.

Most of the registered extensions within this paper's scope are now complete and reported above: the full-panel Fissler–Ziegel $(\mathrm{VaR}, \mathrm{ES})$ re-scoring (Section 6, accuracy layer lowest at both levels; the deployed shift's 2.5% concession quantified there), the 2008 stress window (the 2000–2013 holdout and its battery re-run), and the certainty-equivalent capital translation (Section 7), and the horizon and ES comparison on the 2000–2013 era is complete with a mixed verdict, reported in Section 6: accuracy ties out of era and the $\sqrt{h}$ ES calibration verdict reverses. The signal-absorption study is also complete (Section 5): a state-reweighted historical simulation does not capture the frontier edge, and the failure mode — effective-sample collapse under sharp univariate conditioning — is why the pooled learner is a precondition. Still open: the intraday frontier once tick data is available. Natural extensions beyond this paper: likelihood-free posteriors for Hawkes and jump-diffusion simulators, and vector quantile regression (Carlier–Chernozhukov–Galichon) as the principled multivariate transport map to succeed the scale/shape hybrid.

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Data and code availability

A public replication package reproducing every table and figure in this paper — estimation code for all stages of the engine ((2)–(5)), the misspecification score (9), the FRTB battery, the generative-posterior and SBC routines, and the exact walk-forward, seeds, and hyperparameter grids referenced in the text — is public at `github.com/sean-qin-usa/downside-risk-paper`; a citable archival snapshot (Zenodo DOI) will be minted for the accepted version. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author's institutional licenses and cannot be redistributed; the package includes the code to rebuild every panel from those sources, the derived non-proprietary series, and a synthetic example dataset (`toy_example.py`) that runs the full pipeline end-to-end without licensed data. The pre-registered tuning grid is in `TUNING_GRID_PREREG.md`; the analyses deferred in earlier drafts and since completed (the full-panel Fissler–Ziegel re-scoring, the 2000–2013 holdout and its battery re-run) ship with their result files, and the remaining open items (Section 8) are tracked as issues in the repository.

References

August A. Balkema and Laurens de Haan. Residual life time at great age. *Annals of Probability*, 2(5):792–804, 1974.

Giovanni Barone-Adesi, Kostas Giannopoulos, and Les Vosper. VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5):583–602, 1999.

Basel Committee on Banking Supervision. Minimum capital requirements for market risk. Technical report, Bank for International Settlements, 2019.

Pier Giovanni Bissiri, Chris C. Holmes, and Stephen G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130, 2016.

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

Victor Chernozhukov, Iván Fernández-Val, and Alfred Galichon. Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125, 2010.

Peter F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4): 841–862, 1998.

Will Dabney, Georg Ostrovski, David Silver, and Rémi Munos. Implicit quantile networks for distributional reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1096–1105, 2018.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*. Springer, 1997.

Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

Robert F. Engle and Simone Manganelli. CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381, 2004.

Tobias Fissler and Johanna F. Ziegel. Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4):1680–1707, 2016.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, pages 1660–1672, 2021.

Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5):1779–1801, 1993.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

Peter Reinhard Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005.

Wenxin Jiang and Martin A. Tanner. Gibbs posterior for variable selection in high-dimensional classification and data mining. *Annals of Statistics*, 36(5):2207–2231, 2008.

Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

Paul H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84, 1995.

Alexander J. McNeil and Rüdiger Frey. Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4):271–300, 2000.

Maria Nareklishvili, Nicholas G. Polson, and Vadim Sokolov. Generative quantile Bayesian prediction. *arXiv preprint arXiv:2510.21784*, 2025.

Andrew J. Patton, Johanna F. Ziegel, and Rui Chen. Dynamic semiparametric models for expected shortfall (and Value-at-Risk). *Journal of Econometrics*, 211(2):388–413, 2019.

James Pickands. Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1):119–131, 1975.

Nicholas G. Polson and Vadim Sokolov. Generative AI for Bayesian computation. *arXiv preprint arXiv:2305.14972*, 2023.

Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 3538–3548, 2019.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Chao Zhang, Yihuang Zhang, Mihai Cucuringu, and Zhongmin Qian. Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, 22(2): 492–530, 2024.

Gilles O. Zumbach. The RiskMetrics 2006 methodology. Technical report, RiskMetrics Group, 2007.

A Proofs

The four formal results of Section 3.4 are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

Proof of Lemma 1 (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1-\tau)\int_{(-\infty,q]}(q-z)\,dF(z) \;+\; \tau\int_{(q,\infty)}(z-q)\,dF(z)$$

is finite for every q, and S is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z-q)$). For fixed z, $q \mapsto \rho_\tau(z-q)$ is Lipschitz with constant $\max(\tau, 1-\tau)$, so the difference quotients $[\rho_\tau(Z-q') - \rho_\tau(Z-q)]/(q'-q)$ are bounded by the integrable constant $\max(\tau, 1-\tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of F,

$$\frac{d}{dq}\,\rho_\tau(z-q) = \mathbb{I}\{z < q\} - \tau, \qquad \Longrightarrow \qquad S'(q) = \mathbb{E}\,\mathbb{I}\{Z < q\} - \tau = F(q^-) - \tau,$$

which equals $F(q) - \tau$ for all but countably many q. Since S is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\,du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on F is needed for this identity.* For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is ≥ 0 on $[q_F, q]$, so the integral is ≥ 0; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\, du = \int_{q}^{q_F} \big(\tau - F(u)\big)\, du \;\geq\; 0,$$

the integrand again ≥ 0 on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. u between q_F and q. Finally, if F is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining q_F and q, then F is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^{u} f$; hence $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$, and integrating over the interval yields $\frac{m}{2}(q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q - q_F)^2$. $\square$

Proof of Proposition 1 (split-conformal validity)

Let $k = \lceil (n{+}1)\tau \rceil$ and assume $1 \leq k \leq n$ so that the k-th order statistic $e_{(k)} = c_\tau$ of the n calibration errors $\{e_1, \dots, e_n\}$ exists; when $k = n + 1$ (i.e. τ within $1/(n + 1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n + 1$ errors $e_1, \dots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$) are exchangeable and almost surely distinct. Then the rank R of e_{n+1} among the pooled $n + 1$ values is uniformly distributed on $\{1, \dots, n + 1\}$. By construction $\widetilde{q}(\tau \mid s_{n+1}) = \widehat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \leq c_\tau\} = \{e_{n+1} \leq e_{(k)}\} = \{R \leq k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all e_i are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr\big(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\big) = \Pr(R \leq k) = \frac{k}{n + 1} = \frac{\lceil (n{+}1)\tau \rceil}{n + 1}.$$

Since $(n{+}1)\tau \leq \lceil (n{+}1)\tau \rceil < (n{+}1)\tau + 1$, dividing by $n + 1$ gives $\tau \leq k/(n + 1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on s_{n+1}, because a single scalar shift c_τ pooled across states cannot deliver state-conditional coverage. $\square$

Proof of Proposition 2 (tail wedge)

Fix $\nu > 2$, so the t_ν law has finite variance and its unit-variance version $Z = T_\nu \sqrt{(\nu - 2)/\nu}$ is well defined. The t_ν density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \to \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1 + o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ with $c_\nu = \tilde{c}_\nu((\nu - 2)/\nu)^{\nu/2}$, and leaves the index ν unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1 + o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$, i.e.

$Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}}\,(1 + o(1)) = \text{const} \cdot \tau^{-(1/\nu_1 - 1/\nu_2)}(1 + o(1)) \;\longrightarrow\; \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small τ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes *existence* of τ^*; the actual crossover τ_c where the ordering flips is not claimed here and is located numerically in Remark 1.) Under local truth t_{ν_1}, applying (8) with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with m the minimum of the t_{ν_1} density on the wedge interval. $\square$

Table 1: Summary of results and scope conditions. Edges are percentage pinball reductions unless noted; DM statistics are per-date (Section 5 conventions note), one-sided. Panel A: the deployed amortized residual-hybrid engine (worst case anywhere: a tie). Panel B: boundary evidence from configurations not deployed, which motivates the design.

Panel A: the engine's record Result	Magnitude	Significance	Status
vs GARCH-t / EWMA / HS, h=1 (avg)	+0.3 / +1.5 / +1.9%	DM 4.4–8.3	win
Top misspecification decile	+2.7%	DM 6.5	win (concentrated)
Hyperinflation FX (ARS, EGP)	+12–14%	DM 2.9–3.8	win
Frontier equity indices (4 of 8)	∼+1–3%	DM 2.1–3.5	win
Ten-day horizon vs $\sqrt{h}$-scaling	+1.4% (2014–24)	holds in 2020–22 stress; 2000–13 rerun: tie, DM 0.2	win (era-dependent)
2000–2013 frozen-spec holdout (untouched era)	top decile +0.9%	DM 2.2, registered predictions; frozen-rule restatement in text	win (confirmatory)
Point-in-time universe (Q1-2000 top 200, delistings kept)	top bucket +2.9%; overall +0.6%	DM 6.6 / 9.0, frozen real-time thresholds	win (survivorship armor)
Strict calendar splits (2020 cut; pre-GFC 2007 cut)	top decile +2.5%; overall +0.3%	DM 6.2; 3.1 — no calendar overlap anywhere	win (leakage armor)
Familywise error over the 3 × 10 signal-decile grid	9 cells survive 5% FWER	Romano–Wolf step-down, incl. all three top deciles	win (multiplicity-priced)
Exception battery through the 2008 crisis	Kupiec 94% / Christof. 95% at 99%	date-clustered t 0.17 at 99%; at 97.5%, 0.35 (EVT tail) and −1.09 (deployed shift, conservative); HS, EWMA fail	win (stress)
Joint (VaR, ES) FZ0 score, full panel	accuracy layer lowest at both levels	DM 4.1–4.7 vs GARCH-t, FHS; 6.0–9.2 vs GAS; deployed shift concedes 2.5% (DM −2.0)	win (accuracy layer; the shift trades score for coverage)
$\mathrm{ES}_{97.5}$ over-statement removed	8–11%	calibration, Table 7	win (capital)
Cold start / young listings	+6–10%	full-scale panel	win (unique capability)
Calm-quintile edge	≈ 0, never negative	—	tie (costless to hold)
CAViaR (strongest academic rival)	tie	DM 0.59	tie (co-best)
Panel B: boundary evidence (not the engine) — why the design is what it is			
Bare standalone neural net, single names	−4.5 to −6.2%	DM +12 to +18	why the engine is a hybrid
Standalone nonparametric, calm single assets	GARCH as good or better	pooled DM −9.5 to −13.7	scope: shape edge needs misspecification
Korea 2026 crisis, 23 assets (negative control)	GARCH wins all	DM −2.5 to −5	validates the frontier: price crash ≠ shape break
mk_{63}-reweighted FHS (absorber probe)	worse than plain FHS	DM −11 overall and top decile	why pooling: conditioning alone collapses the sample
Baltic freight; IG credit[a]	+15%; +52%	DM 9.6; 30.6	accuracy only; calibration open

[a] Credit and freight fail breach-independence (exceptions cluster) and are excluded from the win column pending repair. The credit magnitude is from the FRTB-grade single-asset battery; the smaller 4% figure in Table 6 is from the separate lean hybrid-vs-GARCH sweep — different battery, both reported.

Table 2: Model selection by task and regime. Each row is established in the section cited.

Task / regime	Winner	Where
Point quantile accuracy, tabular state	GBM (trees)	§3.3
Single asset, calm ($\sim$90% of asset-days)	GARCH-t	§5
Single asset, top misspecification decile	Nonparametric (+2.7%, DM 6.5)	§5
Amortized cross-name panel, any regime	Nonparametric always; no gate	§5.3
Regulatory ES with a $\geq$GARCH floor	Residual-hybrid + EVT + conformal	§6

Table 3: Sample flow across evaluation panels.

Panel	Source	Era	Names	Test obs	Role
Design equity panel	CRSP/WRDS	2014–2024	200	221,600	frontier map; engine development
Confirmatory holdout	CRSP/WRDS	2000–2013	200	280,608	frozen-spec replication
Exception restatement	CRSP/WRDS	2014–2024	140	≈155,000	per-asset backtesting battery
Cross-asset sweep	Bloomberg	through 2026	43	[a]	frontier breadth
Country indices	Bloomberg	through 2026	26	[a]	frontier breadth

[a] Frozen exhibits; per-instrument windows vary and are documented instrument-by-instrument in the repository.

Table 4: The misspecification frontier: nonparametric edge over GARCH-t by decile of the trailing residual-shape score (200 CRSP names, 221.6k test asset-days). Edge = (GARCH pinball − nonparam pinball)/GARCH pinball.

	Edge by signal decile (%)		
Decile	mk_{63} (kurtosis)	$\lvert\mathrm{sk}_{63}\rvert$ (asymmetry)	jump_5
1–8 (bulk)	≈ 0	≈ 0	mixed, ≈ 0
9	+0.52	+0.69	—
10 (top)	**+2.46**	**+2.37**	+0.90
Overall		+0.44	

Per-date Diebold–Mariano: top mk_{63} decile edge +2.71%, DM 6.54 ($p \approx 0$); bottom decile +0.19%, DM 1.41 ($p = 0.079$, not significant); top two deciles pooled +1.65%, DM 6.74; overall +0.44%, DM 5.67. Top-decile signal means: kurtosis ≈ 25, $\lvert\text{skew}\rvert \approx 4$.

Table 5: Double sort: the edge needs both a shape break and an active regime. Cells are the nonparametric-over-GARCH-t pinball edge (%) in independent quintiles of the misspecification score (rows) and trailing volatility (columns), design-era test panel.

	rv_{63} Q1 (calm)	Q2	Q3	Q4	Q5 (active)
mk_{63} Q1 (low)	+0.24	+0.34	+0.10	+0.01	−0.14
Q2	+0.16	+0.20	+0.22	+0.09	−0.57
Q3	−0.02	+0.19	+0.25	+0.44	+0.04
Q4	+0.08	+0.15	+0.35	+0.23	+0.10
Q5 (high)	−0.10	+0.44	+0.68	+0.61	+**2.33**

Top-row date-clustered DM by volatility quintile: 0.7, 2.3, 1.2, 2.8, 6.4. The bottom-left and top-right regions are the point: neither ingredient alone produces the concentration.

Table 6: The frontier across universes. Ratio = nonparam/GARCH-t pinball (< 1: nonparametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM −13.7
	EM (10)	1.013	8	GARCH, pooled DM −9.5
	crisis (7)	0.973	∼1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86, $p = .002$
	USDEGP	0.857	1920	DM 3.80, $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins[a]
Cross-asset (43)	vol indices	∼0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
13/43 significant wins overall; corr(log kurt, edge) = 0.43				
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM −2.5 to −5

[a] Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06). Cross-index corr(log residual kurtosis, edge) = 0.53. Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating necessity-not-sufficiency.

Table 7: FRTB battery: average pinball loss (twelve quantile levels), $ES_{97.5}$ calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	$ES_{97.5}$ pred. vs realized	$Breach_{99}$	$Kupiec_{99}$ p
Ideal value	*(smaller better)*	*(pred. = realized)*	*1.00%*	> 0.05 *(pass)*
Residual-hybrid (GBM)	**0.3551**	$\mathbf{-5.80}$ / $\mathbf{-5.76}$	1.19%	0.00
+ EVT tail	0.3554	−6.08 / −5.86	**1.04%**	**0.10**
GARCH(1,1)-t	0.3561	−6.17 / −5.65	1.06%	0.026
GARCH-FHS	0.3565	−6.20 / −5.59	1.05%	0.041
GJR-GARCH-skew-t	0.3562	−6.10 / −5.84	1.05%	0.062
EWMA (RiskMetrics)	0.3606	—	—	—
Historical simulation	0.3619	—	—	—

Diebold–Mariano, hybrid vs: GARCH-t 4.38, FHS 5.70, GJR 4.52 (corrected; see below), HS 7.34, EWMA 8.33 (all $p \approx 0$). Implementation note, from adversarial review: the original GJR-skew-t quantile silently reduced to a symmetric $t(8)$ — the fitted skewness and tail parameters were estimated but never applied. The row above is re-estimated with the correct Hansen skew-t inverse CDF (validated against the symmetric-t limit and `arch`'s own distribution; median fitted $\eta = 4.2$, $\lambda = -0.015$). The correction improves the benchmark (pinball 0.3574 → 0.3562; it now passes both 99% exception tests, Kupiec $p = 0.062$, Christoffersen $p = 0.103$) and tightens the hybrid's margin (DM 5.73 → 4.52); a full battery re-run reproduces every other row to the fourth decimal, and the MCS still contains the hybrid alone. Hybrid+EVT still beats GARCH-t (DM 3.90) and FHS (DM 4.28); its 0.0003 pinball concession to the raw hybrid (DM −5.06) buys the exception-test pass. Hybrid+EVT passes $Christoffersen_{99}$ ($p = 0.20$); no other model in the battery passes both 99% tests while topping the accuracy table. At 97.5% *every* raw model fails Kupiec; the split-conformal recalibration repairs it (GBM-recal $p = 0.24$; IQN-v2-recal $p = 0.15$, Christoffersen $p = 0.068$).

Reading guide: $Breach_{99}$ is the realized frequency of losses beyond the 99% VaR — ideal is exactly 1.00% (higher = risk understated, lower = capital wasted); the ES column compares predicted to realized average exceedance loss, ideal is equality; Kupiec p above 0.05 means the breach count is statistically consistent with 1%. An honesty note on scope: the exception statistics here pool the panel's asset-days into a single sequence, so they are *panel calibration diagnostics*, not desk-level regulatory backtests — cross-sectional exceptions on a common date are dependent, and Basel's exception counts apply to a desk's P&L time series. The restatement has now been run (Figure 7): per asset, the EVT-tailed engine passes Kupiec at 99% for 81% of names and Christoffersen for 86%, and on the dependence-respecting date-clustered test (per-date breach frequency against nominal, Newey–West t) it passes at both levels ($t = 1.01$ at 99%, 1.86 at 97.5%), while the raw hybrid is rejected ($t = 4.5$) — the tail stages, not the body, carry the regulatory calibration. The "realized" ES comparator is model-dependent by construction (each model's own breaches select the conditioning set), which is why the joint Fissler–Ziegel score of the preceding paragraph, not this column, is the ranking criterion for ES. Bold marks the best value in each column. "—" entries were not recorded in the archived runs for the lower-accuracy tier (their exception statistics did not affect any ranking); completing them is scheduled with the registered battery re-run.